

The Beast and MILLPAD

THE BEAST AND MILLPAD

A thirty-day fretboard navigation programme for four-string bass

What this manual is. A complete, self-contained system for learning to navigate the entire bass fretboard using one exercise (The Beast) and one organising principle (MILLPAD). It assumes you can play a bass and read tablature, and nothing else.

What it is not. A collection of tab to memorise. If at the end of thirty days you can play the C-major traversal beautifully but cannot generate the same thing in E-flat from a random fret, the manual has failed and so has the practice.

How the material is arranged.

Part	Contents	When to use it
One	The connection — why these two things are one thing	Read completely before Day 1
Two	Reference atlas — every chart, every key	Look things up; never read cover to cover
Three	The thirty-day programme, one hour per day	One day at a time, in order
Four	Twenty-one exploration passages	As directed by Part Three
Five	Troubleshooting, route cards, logs	When something breaks

A note on the source material. This system comes from Josh Fossgreen’s presentation of The Beast and Anthony Wellington’s MILLPAD organisation. Widely circulated transcriptions of both contain errors in the shape definitions, which make a consistent system look self-contradictory. Section 1.8 lists the corrections. If something here disagrees with something you have read elsewhere, check that section first.

Safety. The WS-WS shape spans five frets. In the lower register you should **shift the hand**, not stretch. Stop immediately on sharp pain, tingling or numbness. No fretboard knowledge is worth a hand injury, and there is no position on the neck where the correct fingering hurts.

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PART ONE — THE CONNECTION

1.1 Read this chapter before you touch the bass

You are not confused because you are slow. You are confused because the material is almost always taught as two unrelated things:

- **The Beast** is presented as a *route*: a long snake of notes up and down the neck.
- **MILLPAD** is presented as a *word*: seven mode names in a strange order.

Nobody tells you the one sentence that joins them:

MILLPAD is not a second exercise. MILLPAD is the label track for The Beast. The Beast moves your hand. MILLPAD tells you what your hand is about to do next.

Everything in this manual is an unpacking of that sentence. By the end of this chapter you should be able to explain, out loud and without the bass, *why* the seven-letter word exists, *why* it is in that order, *why* there are only three hand shapes, and *why* there is exactly one fret shift in the entire system.

If you can explain those four things, the remaining twenty-nine days are just drilling.

1.2 What The Beast actually is (stripped of mystique)

Take one seven-note scale. Play it in a straight line, low to high, never repeating a note, never skipping a note. Now impose one single rule:

Three notes per string. Then move to the next string.

That is the entire Beast. Everything else — the turnarounds, the climbing, the modal labels — falls out of that one rule automatically.

On a four-string bass, “three per string” times “four strings” gives you **twelve notes per crossing**. Twelve consecutive scale notes. That block of twelve is called a **sweep**.

Here is the first sweep in C major from the open E string:

Scale, written as an unbroken line: E F G | A B C | D E F | G A B
Assigned three at a time to strings: E string | A string | D string | G string

```
G|-----0--2--4-|
D|-----0--2--3-|
A|-----0--2--3-|
E|---0--1--3-----|
```

Look at what happened. You did not “choose” any of those frets. You wrote the scale in a line, cut it into fours, and handed each group to the next string. **The tab is a consequence, not a decision.**

That is the first thing to internalise: *The Beast is not a shape you memorise. It is a rule you apply.*

The turnaround

When you run out of strings you have two choices: jump back down and repeat the same twelve notes forever, or **keep the scale moving**. The Beast keeps it moving. You take **one more scale note** — the next one above where you stopped — and then reverse.

The first sweep ended on B at G-string fret 4. The next C-major note above it is C at fret 5. So you play C, and then descend:

```
G|---5--4--2-----|
D|-----5--3--2-----|
A|-----5--3--2-----|
E|-----5--3--1-----|
```

That extra note is the hinge. It is why the exercise climbs the neck instead of running in a loop. Each complete up-and-down cycle starts **two scale notes higher** than the last one.

The two-note rule. Sweep 1 (up) starts on scale note 1. Sweep 3 (up) starts on scale note 3. Sweep 5 (up) starts on scale note 5. Up two, every time. This is the only piece of arithmetic in the whole system, and you will use it to check yourself when you get lost.

1.3 The hinge: three notes on a string tuned in fourths

Now the actual connection. This is the section to reread until it is obvious.

Your bass is tuned in **perfect fourths**: E to A is five semitones, A to D is five semitones, D to G is five semitones. Every string crossing is worth exactly **+5 frets of pitch**.

The Beast puts **three scale notes** on each string. So when you cross to the next string, the note you start on is **three scale steps higher** than the note you started the previous string on.

Hold those two facts next to each other:

	Amount
One string crossing, in pitch	+5 semitones (fixed by tuning)
Three diatonic steps, in pitch	+5 semitones — usually

Three steps up a major scale is a **fourth**. C to F. D to G. E to A. Almost always five semitones. Which means:

Crossing a string and advancing three scale notes cancel each other out. Your hand stays in the same fret column.

That is why The Beast feels like a box and not like a climb. Look again at the first sweep: frets 0-1-3 / 0-2-3 / 0-2-3 / 0-2-4. Every single group starts on fret 0. Four strings, twelve notes, one hand position.

Where it breaks — and why that is the good part

Three diatonic steps is a perfect fourth **six times out of seven**. Once per octave it is a **tritone** — six semitones, not five. In C major that one exception is **F up to B**.

The tuning gives you 5. The music demands 6. You are one semitone short. So on that one crossing — and *only* that one — **your hand moves up one fret**.

```
F -> B    is 6 semitones (tritone)
string    gives 5 semitones
-----
shortfall = 1 fret  ->  slide the hand up one fret, once per cycle
```

That single fret is the entire reason The Beast climbs the neck at all. Everything else in the system is stationary; this one crossing is the ratchet.

1.4 MILLPAD is the name of that seven-step cycle

We now have a cycle: *start a group, advance three scale notes, start the next group*. Advance three notes seven times and you are back where you started ($3 \times 7 = 21 =$ three octaves). So the cycle has **exactly seven positions** and then repeats forever.

Let us walk it in C major, starting on G:

```
G  ->  C  ->  F  ->  B  ->  E  ->  A  ->  D  ->  G  ...
+3    +3    +3    +3    +3    +3    +3    (scale steps)
4th   4th   TRITONE 4th   4th   4th   4th   (actual interval)
```

Now label each of those notes with the mode that begins on it in C major:

G	C	F	B	E	A	D
Mixolydian	Ionian	Lydian	Locrian	Phrygian	Aeolian	Dorian
M	I	Ly	Lo	P	A	D

M - I - Ly - Lo - P - A - D.

That is MILLPAD. It is not an arbitrary mnemonic and it is not a list of modes to practise. **It is the order in which three-note groups arrive under your hand when you play three notes per string on an instrument tuned in fourths**. It could not have been any other order.

The one-sentence connection. The Beast advances three scale notes per string. Three scale notes is a fourth. Seven fourths closes a circle. MILLPAD is that circle, written as mode names.

And notice where the tritone landed: between **Ly and Lo**. Remember that. It is the only place in the entire system where anything moves.

1.5 Why there are only three hand shapes

A three-note group's physical shape depends on only one thing: the two intervals inside it. In a major scale, semitones sit between degrees 3-4 and 7-1. So a three-note group can contain the semitone in the first gap, the second gap, or neither. Three possibilities. **There cannot be a fourth.**

MILLPAD	Mode	Degree in parent key	First two steps	Shape	Fret offsets	Fingers
M	Mixolydian	5	W-W	WS-WS	0, +2, +4	1-3-4 high / 1-2-4 low + shift
I	Ionian	1	W-W	WS-WS	0, +2, +4	1-3-4 high / 1-2-4 low + shift
Ly	Lydian	4	W-W	WS-WS	0, +2, +4	1-3-4 high / 1-2-4 low + shift
Lo	Locrian	7	H-W	1-2-4	0, +1, +3	1-2-4
P	Phrygian	3	H-W	1-2-4	0, +1, +3	1-2-4
A	Aeolian	6	W-H	1-3-4	0, +2, +3	1-3-4
D	Dorian	2	W-H	1-3-4	0, +2, +3	1-3-4

Written as shapes only:

```
WS-WS  x, x+2, x+4  W-W  "wide"      M I Ly
1-2-4   x, x+1, x+3  H-W  "half first" Lo P
1-3-4   x, x+2, x+3  W-H  "half last"  A D
```

Which gives the master grid — the thing to tattoo on the inside of your eyelids:

```
  M   I   Ly | Lo  P  | A   D
W-W W-W W-W | H-W H-W| W-H W-H
<--- three ---> <- two -> <- two ->
```

Three wide, two half-first, two half-last. Say it as a rhythm: *three - two - two*.

Why do they clump so neatly? Because MILLPAD orders the degrees by fourths, and ordering by fourths sorts the two semitones of the scale into adjacent slots. The clean 3-2-2 block structure is a *consequence* of the fourths ordering, which is itself a consequence of the three-notes-per-string rule. The whole system is one idea wearing three costumes.

1.6 The single shift, stated precisely

Put the two discoveries together:

1. Crossing a string while advancing three scale notes keeps your hand in the same fret column.
2. Except at **Ly to Lo**, where the interval is a tritone and the hand moves **up one fret**.

So for any ascending sweep, the fret columns are:

E-string label	E string	A string	D string	G string	Shapes E-A-D-G	Fret columns
M	M	I	Ly	Lo	W-W W-W W-W H-W	x x x x+1
I	I	Ly	Lo	P	W-W W-W H-W H-W	x x x+1 x+1
Ly	Ly	Lo	P	A	W-W H-W H-W W-H	x x+1 x+1 x+1
Lo	Lo	P	A	D	H-W H-W W-H W-H	x x x x
P	P	A	D	M	H-W W-H W-H W-W	x x x x
A	A	D	M	I	W-H W-H W-W W-W	x x x x
D	D	M	I	Ly	W-H W-W W-W W-W	x x x x

Read the last column. Six of the seven rows are dead flat until they hit the Ly-to-Lo boundary. If your starting label is Lo, P, A or D, the boundary does not occur inside your four strings at all, and **the entire sweep sits in one fret column**.

Memory phrase: *After three wides, the index steps forward.* The three wides are M, I, Ly. The step forward is into Lo.

Descending is MILLPAD backwards

When you turn around at the G string, you are still moving three scale notes per crossing — just downward. So you read the same cycle in reverse:

Forwards (ascending): M -> I -> Ly -> Lo -> P -> A -> D -> M
Backwards (descending): M -> D -> A -> P -> Lo -> Ly -> I -> M

The first C-major descent is labelled **A - P - Lo - Ly** — check it against the backwards cycle. It is exactly four consecutive labels read right to left.

And the shift reverses too: descending, your hand steps **down** one fret when you cross **Lo to Ly**.

Why the descending label looks “wrong”. The first descending group in C major sounds C-B-A but is labelled Aeolian. The label always describes the shape *as it would be written ascending*. Lowest to highest, C-B-A is A-B-C, which is W-H, which is the 1-3-4 / Aeolian shape. You are playing an Aeolian chunk backwards. The label names the shape under your fingers, not the first note you hear.

1.7 The four things you actually track

Twelve fret numbers per sweep is far too much to hold. You will never navigate a fretboard by remembering numbers. You navigate it by holding **four small variables** and regenerating the numbers on the fly:

KEY -- DIRECTION -- STRING -- CHUNK

Example state: C major - ascending - D string - Dorian.

From that alone, everything is recoverable:

- Dorian means the **1-3-4** shape ($x, x+2, x+3$).
- Dorian in C major means the group starts on **D**.
- The next label ascending is **Mixolydian**, on the **G string**.
- Ly to Lo is not the next crossing, so **no fret shift**.

That is the skill. Not the tab. **The tab is disposable; the state model is the instrument.**

1.8 Corrections to the document you were reading

Your source PDF is broadly right but contains transcription errors that make a consistent system look self-contradictory. Fix these in your head now.

Source text	Correct	Why
1-2-4 illustrated as frets 5-7-8	1-2-4 is frets 5-6-8	Fingers 1, 2 and 4 sit on frets $x, x+1, x+3$. Interval H-W.
1-3-4 illustrated as frets 3-5-7	1-3-4 is frets 5-7-8	Fingers 1, 3 and 4 sit on frets $x, x+2, x+3$. Interval W-H.
A rule labelled 2-3-4	There is no such shape	Only three shapes exist. It is a typo for 1-3-4.
“WS-WS is fingered 1-3-4” stated absolutely	Position-dependent	WS-WS spans five frets. Below about fret 7 you shift the hand; above it you may reach.

The reliable self-test: **fingers are named after fingers, not frets**. “1-2-4” means you use fingers one, two and four. Fingers two and four are two frets apart in a one-finger-per-fret hand, so the frets are $x, x+1, x+3$. Do that reasoning once and you will never mix them up again.

1.9 What MILLPAD is *not*

MILLPAD is a **fretboard organisation system**, not a claim about how the music sounds.

Playing E-F-G inside a C-major Beast does not make anyone hear E Phrygian. A mode is a *perceived tonal centre*, and a tonal centre requires harmonic and rhythmic emphasis — a drone, a bass pedal, a chord, a phrase that resolves. Three

passing notes cannot do that.

So hold two ideas separately and permanently:

- “**Phrygian chunk**” = a physical shape (H-W) that starts on the third degree of the parent key.
- “**E Phrygian**” = music in which E is heard as home and the C-major collection is heard around it.

Days 27 to 30 of this programme are where you deliberately convert the first into the second. Until then, treat the mode names as **filenames for hand shapes** and you will stay sane.

1.10 A worked generation, start to finish

Here is the whole system used once, in a key not covered anywhere in Part Three, with every step written out. Nothing is looked up. If you can follow this, you can do it in E major on Day 24 without ever having practised E major.

Task: start a sweep in **E major**, from the note on the E string at **fret 4**.

Step 1 — spell the key. E major has four sharps: **E F# G# A B C# D#**.

Step 2 — number the degrees. E=1, F#=2, G#=3, A=4, B=5, C#=6, D#=7.

Step 3 — identify the starting note. E string, fret 4. Open E, up four semitones: E, F, F#, G, **G#**. So the note is G#.

Step 4 — degree and label. G# is degree **3**. Degree 3 is **Phrygian**, label **P**.

Step 5 — read four labels forwards. MILLPAD is M I Ly Lo P A D. Starting at P: **P - A - D - M**.

Step 6 — shapes. P is 1-2-4 (H-W). A is 1-3-4 (W-H). D is 1-3-4 (W-H). M is WS-WS (W-W).

Step 7 — fret columns. The Ly-to-Lo crossing does not appear in P-A-D-M, so **there is no shift**. All four groups start at fret 4.

Step 8 — write it.

E string	P	1-2-4	4-5-7	G#	A	B
A string	A	1-3-4	4-6-7	C#	D#	E
D string	D	1-3-4	4-6-7	F#	G#	A
G string	M	WS-WS	4-6-8	B	C#	D#

Step 9 — check the music, not the shapes. Written as an unbroken line the twelve notes are G# A B C# D# E F# G# A B C# D#. That is the E-major scale running without gaps or repeats, which is the only real test. If your notes do not form an unbroken scale, the tab is wrong no matter how good the shapes look.

Step 10 — the turn. The sweep ends on D#. The next E-major note above it is **E**, on the G string at fret 9. Play it, then descend, reading the labels backwards: **M - D - A - P**.

Total elapsed time once this is fluent: about five seconds. That is the skill the thirty days are buying.

1.11 Chapter self-test

Do not turn the page until you can answer all nine without looking. Write the answers.

1. State the single rule that generates The Beast.
2. How many notes are in one four-string sweep, and why that number?
3. Why does crossing a string usually *not* change your fret column?
4. Name the one interval that breaks the pattern, and where it occurs in MILLPAD.
5. How many hand shapes exist, and why can there not be a fourth?
6. Recite the master grid as “three - two - two” with the interval of each.
7. Given the state **G major - ascending - A string - Locrian**, what shape are you playing, what is the next label, and does your hand shift?
8. Why is a descending group that sounds C-B-A labelled Aeolian?
9. What does each complete up-and-down cycle do to your starting scale note?

(Answers: question 8 is the only one with any subtlety — the descending label names the ascending shape. If you missed 3, 4 or 8, reread 1.3, 1.4 and 1.6 before starting Day 1.)

PART TWO — REFERENCE ATLAS

Do not read this part. Use it. Every chart here is generated from the same rules described in Part One, so if you ever disagree with a chart, work the rule and see which of you is wrong.

2.1 The neck

Every note, frets 0 to 12. Above fret 12 everything repeats an octave higher.

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	G	G#/Ab	A	A#/Bb	B	C	C#/Db	D	D#/Eb	E	F	F#/Gb	G
D	D	D#/Eb	E	F	F#/Gb	G	G#/Ab	A	A#/Bb	B	C	C#/Db	D
A	A	A#/Bb	B	C	C#/Db	D	D#/Eb	E	F	F#/Gb	G	G#/Ab	A
E	E	F	F#/Gb	G	G#/Ab	A	A#/Bb	B	C	C#/Db	D	D#/Eb	E

Natural notes only. This is the skeleton worth memorising; everything else is a natural note plus or minus one fret.

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	G	.	A	.	B	C	.	D	.	E	F	.	G
D	D	.	E	F	.	G	.	A	.	B	C	.	D
A	A	.	B	C	.	D	.	E	F	.	G	.	A
E	E	F	.	G	.	A	.	B	C	.	D	.	E

Two facts generate the whole table:

- **B to C and E to F are one fret apart.** Every other adjacent letter pair is two frets.
- The strings are tuned in perfect fourths, so **fret 5 on any string is the next open string.**

2.2 Anchors and octaves

Learn landmarks, not fifty-two independent cells.

Fret	Function	E - A - D - G
0	String names	E, A, D, G
5	Same pitch as the next open string	A, D, G, C
7	Secondary landmark	B, E, A, D
12	Octave of the open string	E, A, D, G

The octave shape. From any note on the E or A string, its octave is **two strings up and two frets across**.

E string fret 3 = G	D string fret 5 = G
A string fret 3 = C	G string fret 5 = C
E string fret 5 = A	D string fret 7 = A

This follows directly from fourths tuning: two strings up is ten semitones, plus two frets is twelve, which is an octave. It is not a coincidence and it is worth deriving once.

2.3 The shape atlas

Three shapes. There cannot be a fourth, because a three-note group drawn from a major scale either contains the semitone first, or second, or not at all.

WS-WS	x, x+2, x+4	W-W	span 5 frets	M I Ly
1-2-4	x, x+1, x+3	H-W	span 4 frets	Lo P
1-3-4	x, x+2, x+3	W-H	span 4 frets	A D

Shape	Frets from 5	From 7	From 9	From 12	Fingering
WS-WS	5-7-9	7-9-11	9-11-13	12-14-16	1-3-4 above fret 7; shift the hand below it
1-2-4	5-6-8	7-8-10	9-10-12	12-13-15	1-2-4
1-3-4	5-7-8	7-9-10	9-11-12	12-14-15	1-3-4

Open-string forms. Treat an open string as an imaginary first finger. The interval pattern is unchanged; only the physical fingering differs.

WS-WS open: 0-2-4 1-2-4 open: 0-1-3 1-3-4 open: 0-2-3

2.4 The master grid

MILLPAD	Mode	Degree in parent key	First two steps	Shape	Fret offsets	Fingers
M	Mixolydian	5	W-W	WS-WS	0, +2, +4	1-3-4 high / 1-2-4 low + shift
I	Ionian	1	W-W	WS-WS	0, +2, +4	1-3-4 high / 1-2-4 low + shift
Ly	Lydian	4	W-W	WS-WS	0, +2, +4	1-3-4 high / 1-2-4 low + shift
Lo	Locrian	7	H-W	1-2-4	0, +1, +3	1-2-4
P	Phrygian	3	H-W	1-2-4	0, +1, +3	1-2-4
A	Aeolian	6	W-H	1-3-4	0, +2, +3	1-3-4
D	Dorian	2	W-H	1-3-4	0, +2, +3	1-3-4

M I Ly | Lo P | A D
W-W W-W W-W | H-W H-W | W-H W-H
<--- three ---> <- two -> <- two ->

Forwards for ascending, backwards for descending:

UP M -> I -> Ly -> Lo -> P -> A -> D -> M ... shift +1 at Ly->Lo
DOWN M -> D -> A -> P -> Lo -> Ly -> I -> M ... shift -1 at Lo->Ly

2.5 The seven starting rows

E-string label	E string	A string	D string	G string	Shapes E-A-D-G	Fret columns
M	M	I	Ly	Lo	W-W W-W W-W H-W	x x x x+1
I	I	Ly	Lo	P	W-W W-W H-W H-W	x x x+1 x+1
Ly	Ly	Lo	P	A	W-W H-W H-W W-H	x x+1 x+1 x+1
Lo	Lo	P	A	D	H-W H-W W-H W-H	x x x x
P	P	A	D	M	H-W W-H W-H W-W	x x x x
A	A	D	M	I	W-H W-H W-W W-W	x x x x
D	D	M	I	Ly	W-H W-W W-W W-W	x x x x

Four of the seven rows — Lo, P, A and D — contain no shift at all, because the Ly-to-Lo boundary falls outside their four strings. Those are the rows to learn first.

2.6 The twelve keys

Each key below gives: the spelled notes, the scale notes on the neck, the MILLPAD label map, a summary of the traversal, and the tab for the first four sweeps from the lowest playable position.

Use these to check yourself, never to learn from. Generate the row on paper using the seven-step procedure in Phase 5, then look here.

C major

Notes: C D E F G A B

Scale notes on the neck (frets 0-12):

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	G	.	A	.	B	C	.	D	.	E	F	.	G
D	D	.	E	F	.	G	.	A	.	B	C	.	D
A	A	.	B	C	.	D	.	E	F	.	G	.	A
E	E	F	.	G	.	A	.	B	C	.	D	.	E

Same neck, MILLPAD label of the chunk that STARTS on each note:

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	M	.	A	.	Lo	I	.	D	.	P	Ly	.	M
D	D	.	P	Ly	.	M	.	A	.	Lo	I	.	D
A	A	.	Lo	I	.	D	.	P	Ly	.	M	.	A
E	P	Ly	.	M	.	A	.	Lo	I	.	D	.	P

The full traversal, lowest playable start upward:

#	Dir	Labels (play order)	E	A	D	G	Notes (play order)
1	UP	P A D M	0-1-3	0-2-3	0-2-3	0-2-4	E F G A B C D E F G A B
2	DOWN	A P Lo Ly	5-3-1	5-3-2	5-3-2	5-4-2	C B A G F E D C B A G F
3	UP	M I Ly Lo	3-5-7	3-5-7	3-5-7	4-5-7	G A B C D E F G A B C D
4	DOWN	I M D A	8-7-5	8-7-5	9-7-5	9-7-5	E D C B A G F E D C B A
5	UP	Lo P A D	7-8-10	7-8-10	7-9-10	7-9-10	B C D E F G A B C D E F
6	DOWN	P Lo Ly I	12-10-8	12-10-8	12-10-9	12-10-9	G F E D C B A G F E D C

Sweep 1 (ascending): $P > A > D > M$

G	-----0--2--4-
D	-----0--2--3-----
A	-----0--2--3-----
E	---0--1--3-----

Sweep 2 (descending): $A > P > Lo > Ly$

G	---5--4--2-----
D	-----5--3--2-----
A	-----5--3--2-----
E	-----5--3--1-

Sweep 3 (ascending): $M > I > Ly > Lo$

G	-----4--5--7-
D	-----3--5--7-----
A	-----3--5--7-----
E	---3--5--7-----

Sweep 4 (descending): $I > M > D > A$

G	---9--7--5-----
D	-----9--7--5-----
A	-----8--7--5-----
E	-----8--7--5-

Sweep 5 (ascending): $Lo > P > A > D$

G	-----7--9-10-
D	-----7--9-10-----
A	-----7--8-10-----
E	---7--8-10-----

Sweep 6 (descending): $P > Lo > Ly > I$

G	--12-10--9-----
D	-----12-10--9-----
A	-----12-10--8-----
E	-----12-10--8-

G major

Notes: G A B C D E F#

Scale notes on the neck (frets 0-12):

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	G	.	A	.	B	C	.	D	.	E	.	F#	G
D	D	.	E	.	F#	G	.	A	.	B	C	.	D
A	A	.	B	C	.	D	.	E	.	F#	G	.	A
E	E	.	F#	G	.	A	.	B	C	.	D	.	E

Same neck, MILLPAD label of the chunk that STARTS on each note:

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	I	.	D	.	P	Ly	.	M	.	A	.	Lo	I
D	M	.	A	.	Lo	I	.	D	.	P	Ly	.	M
A	D	.	P	Ly	.	M	.	A	.	Lo	I	.	D
E	A	.	Lo	I	.	D	.	P	Ly	.	M	.	A

The full traversal, lowest playable start upward:

#	Dir	Labels (play order)	E	A	D	G	Notes (play order)
1	UP	A D M I	0-2-3	0-2-3	0-2-4	0-2-4	E F# G A B C D E F# G A B
2	DOWN	D A P Lo	5-3-2	5-3-2	5-4-2	5-4-2	C B A G F# E D C B A G F#
3	UP	I Ly Lo P	3-5-7	3-5-7	4-5-7	4-5-7	G A B C D E F# G A B C D
4	DOWN	Ly I M D	8-7-5	9-7-5	9-7-5	9-7-5	E D C B A G F# E D C B A
5	UP	P A D M	7-8-10	7-9-10	7-9-10	7-9-11	B C D E F# G A B C D E F#
6	DOWN	A P Lo Ly	12-10-8	12-10-9	12-10-9	12-11-9	G F# E D C B A G F# E D C

Sweep 1 (ascending): A > D > M > I

G	-----0--2--4-
D	-----0--2--4-----
A	-----0--2--3-----
E	---0--2--3-----

Sweep 2 (descending): D > A > P > Lo

G	---5--4--2-----
D	-----5--4--2-----
A	-----5--3--2-----
E	-----5--3--2-

Sweep 3 (ascending): I > Ly > Lo > P

G	-----4--5--7-
D	-----4--5--7-----
A	-----3--5--7-----
E	---3--5--7-----

Sweep 4 (descending): Ly > I > M > D

G	---9--7--5-----
D	-----9--7--5-----
A	-----9--7--5-----
E	-----8--7--5-

Sweep 5 (ascending): P > A > D > M

G	-----7--9-11-
D	-----7--9-10-----
A	-----7--9-10-----
E	---7--8-10-----

Sweep 6 (descending): A > P > Lo > Ly

G	--12-11--9-----
D	-----12-10--9-----
A	-----12-10--9-----
E	-----12-10--8-

D major

Notes: D E F# G A B C#

Scale notes on the neck (frets 0-12):

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	G	.	A	.	B	.	C#	D	.	E	.	F#	G
D	D	.	E	.	F#	G	.	A	.	B	.	C#	D
A	A	.	B	.	C#	D	.	E	.	F#	G	.	A
E	E	.	F#	G	.	A	.	B	.	C#	D	.	E

Same neck, MILLPAD label of the chunk that STARTS on each note:

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	Ly	.	M	.	A	.	Lo	I	.	D	.	P	Ly
D	I	.	D	.	P	Ly	.	M	.	A	.	Lo	I
A	M	.	A	.	Lo	I	.	D	.	P	Ly	.	M
E	D	.	P	Ly	.	M	.	A	.	Lo	I	.	D

The full traversal, lowest playable start upward:

#	Dir	Labels (play order)	E	A	D	G	Notes (play order)
1	UP	D M I Ly	0-2-3	0-2-4	0-2-4	0-2-4	E F# G A B C# D E F# G A B
2	DOWN	M D A P	5-3-2	5-4-2	5-4-2	6-4-2	C# B A G F# E D C# B A G F#
3	UP	Ly Lo P A	3-5-7	4-5-7	4-5-7	4-6-7	G A B C# D E F# G A B C# D
4	DOWN	Lo Ly I M	9-7-5	9-7-5	9-7-5	9-7-6	E D C# B A G F# E D C# B A
5	UP	A D M I	7-9-10	7-9-10	7-9-11	7-9-11	B C# D E F# G A B C# D E F#
6	DOWN	D A P Lo	12-10-9	12-10-9	12-11-9	12-11-9	G F# E D C# B A G F# E D C#

Sweep 1 (ascending): D > M > I > Ly

```
G|-----0--2--4-|
D|-----0--2--4-----|
A|-----0--2--4-----|
E|---0--2--3-----|
```

Sweep 2 (descending): M > D > A > P

```
G|---6--4--2-----|
D|-----5--4--2-----|
A|-----5--4--2-----|
E|-----5--3--2-|
```

Sweep 3 (ascending): Ly > Lo > P > A

```
G|-----4--6--7-|
D|-----4--5--7-----|
A|-----4--5--7-----|
E|---3--5--7-----|
```

Sweep 4 (descending): Lo > Ly > I > M

```
G|---9--7--6-----|
D|-----9--7--5-----|
A|-----9--7--5-----|
E|-----9--7--5-|
```

Sweep 5 (ascending): A > D > M > I

```
G|-----7--9-11-|
D|-----7--9-11-----|
A|-----7--9-10-----|
E|---7--9-10-----|
```

Sweep 6 (descending): D > A > P > Lo

```

G|--12-11--9-----|
D|-----12-11--9-----|
A|-----12-10--9-----|
E|-----12-10--9-----|

```

A major

Notes: A B C# D E F# G#

Scale notes on the neck (frets 0-12):

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	.	G#	A	.	B	.	C#	D	.	E	.	F#	.
D	D	.	E	.	F#	.	G#	A	.	B	.	C#	D
A	A	.	B	.	C#	D	.	E	.	F#	.	G#	A
E	E	.	F#	.	G#	A	.	B	.	C#	D	.	E

Same neck, MILLPAD label of the chunk that STARTS on each note:

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	.	Lo	I	.	D	.	P	Ly	.	M	.	A	.
D	Ly	.	M	.	A	.	Lo	I	.	D	.	P	Ly
A	I	.	D	.	P	Ly	.	M	.	A	.	Lo	I
E	M	.	A	.	Lo	I	.	D	.	P	Ly	.	M

The full traversal, lowest playable start upward:

#	Dir	Labels (play order)	E	A	D	G	Notes (play order)
1	UP	M I Ly Lo	0-2-4	0-2-4	0-2-4	1-2-4	E F# G# A B C# D E F# G# A B
2	DOWN	I M D A	5-4-2	5-4-2	6-4-2	6-4-2	C# B A G# F# E D C# B A G# F#
3	UP	Lo P A D	4-5-7	4-5-7	4-6-7	4-6-7	G# A B C# D E F# G# A B C# D
4	DOWN	P Lo Ly I	9-7-5	9-7-5	9-7-6	9-7-6	E D C# B A G# F# E D C# B A
5	UP	D M I Ly	7-9-10	7-9-11	7-9-11	7-9-11	B C# D E F# G# A B C# D E F#
6	DOWN	M D A P	12-10-9	12-11-9	12-11-9	13-11-9	G# F# E D C# B A G# F# E D C#

Sweep 1 (ascending): M > I > Ly > Lo

```

G|-----1--2--4-|
D|-----0--2--4-|
A|-----0--2--4-|
E|---0--2--4-----|

```

Sweep 2 (descending): I > M > D > A

```

G|---6--4--2-----|
D|-----6--4--2-----|
A|-----5--4--2-----|
E|-----5--4--2-----|

```

Sweep 3 (ascending): Lo > P > A > D

```

G|-----4--6--7-|
D|-----4--6--7-----|
A|-----4--5--7-----|
E|---4--5--7-----|

```

Sweep 4 (descending): P > Lo > Ly > I

```

G|---9--7--6-----|
D|-----9--7--6-----|
A|-----9--7--5-----|
E|-----9--7--5-----|

```

Sweep 5 (ascending): D > M > I > Ly

```
G|-----7--9-11-|
D|-----7--9-11-|
A|-----7--9-11-|
E|---7--9-10-----|
```

Sweep 6 (descending): $M > D > A > P$

```
G|--13-11--9-----|
D|-----12-11--9-----|
A|-----12-11--9-----|
E|-----12-10--9-----|
```

E major

Notes: E F# G# A B C# D#

Scale notes on the neck (frets 0-12):

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	.	G#	A	.	B	.	C#	.	D#	E	.	F#	.
D	.	D#	E	.	F#	.	G#	A	.	B	.	C#	.
A	A	.	B	.	C#	.	D#	E	.	F#	.	G#	A
E	E	.	F#	.	G#	A	.	B	.	C#	.	D#	E

Same neck, MILLPAD label of the chunk that STARTS on each note:

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	.	P	Ly	.	M	.	A	.	Lo	I	.	D	.
D	.	Lo	I	.	D	.	P	Ly	.	M	.	A	.
A	Ly	.	M	.	A	.	Lo	I	.	D	.	P	Ly
E	I	.	D	.	P	Ly	.	M	.	A	.	Lo	I

The full traversal, lowest playable start upward:

#	Dir	Labels (play order)	E	A	D	G	Notes (play order)
1	UP	I Ly Lo P	0-2-4	0-2-4	1-2-4	1-2-4	E F# G# A B C# D# E F# G# A B
2	DOWN	Ly I M D	5-4-2	6-4-2	6-4-2	6-4-2	C# B A G# F# E D# C# B A G# F#
3	UP	P A D M	4-5-7	4-6-7	4-6-7	4-6-8	G# A B C# D# E F# G# A B C# D#
4	DOWN	A P Lo Ly	9-7-5	9-7-6	9-7-6	9-8-6	E D# C# B A G# F# E D# C# B A
5	UP	M I Ly Lo	7-9-11	7-9-11	7-9-11	8-9-11	B C# D# E F# G# A B C# D# E F#
6	DOWN	I M D A	12-11-9	12-11-9	13-11-9	13-11-9	G# F# E D# C# B A G# F# E D# C#

Sweep 1 (ascending): $I > Ly > Lo > P$

```
G|-----1--2--4-|
D|-----1--2--4-|
A|-----0--2--4-|
E|---0--2--4-----|
```

Sweep 2 (descending): $Ly > I > M > D$

```
G|---6--4--2-----|
D|-----6--4--2-----|
A|-----6--4--2-----|
E|-----5--4--2-----|
```

Sweep 3 (ascending): $P > A > D > M$

```
G|-----4--6--8-|
D|-----4--6--7-----|
A|-----4--6--7-----|
E|---4--5--7-----|
```

Sweep 4 (descending): $A > P > Lo > Ly$

```
G|---9--8--6-----|
D|-----9--7--6-----|
A|-----9--7--6-----|
E|-----9--7--5-----|
```

Sweep 5 (ascending): $M > I > L_y > L_o$

```
G|-----8--9-11-|
D|-----7--9-11-|
A|-----7--9-11-|
E|---7--9-11-----|
```

Sweep 6 (descending): $I > M > D > A$

```
G|--13-11--9-----|
D|-----13-11--9-----|
A|-----12-11--9-----|
E|-----12-11--9-----|
```

B major

Notes: B C# D# E F# G# A#

Scale notes on the neck (frets 0-12):

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	.	G#	.	A#	B	.	C#	.	D#	E	.	F#	.
D	.	D#	E	.	F#	.	G#	.	A#	B	.	C#	.
A	.	A#	B	.	C#	.	D#	E	.	F#	.	G#	.
E	E	.	F#	.	G#	.	A#	B	.	C#	.	D#	E

Same neck, MILLPAD label of the chunk that STARTS on each note:

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	.	A	.	Lo	I	.	D	.	P	L_y	.	M	.
D	.	P	L_y	.	M	.	A	.	Lo	I	.	D	.
A	.	Lo	I	.	D	.	P	L_y	.	M	.	A	.
E	L_y	.	M	.	A	.	Lo	I	.	D	.	P	L_y

The full traversal, lowest playable start upward:

#	Dir	Labels (play order)	E	A	D	G	Notes (play order)
1	UP	L_y Lo P A	0-2-4	1-2-4	1-2-4	1-3-4	E F# G# A# B C# D# E F# G# A# B
2	DOWN	Lo L_y I M	6-4-2	6-4-2	6-4-2	6-4-3	C# B A# G# F# E D# C# B A# G# F#
3	UP	A D M I	4-6-7	4-6-7	4-6-8	4-6-8	G# A# B C# D# E F# G# A# B C# D#
4	DOWN	D A P Lo	9-7-6	9-7-6	9-8-6	9-8-6	E D# C# B A# G# F# E D# C# B A#
5	UP	I L_y Lo P	7-9-11	7-9-11	8-9-11	8-9-11	B C# D# E F# G# A# B C# D# E F#
6	DOWN	L_y I M D	12-11-9	13-11-9	13-11-9	13-11-9	G# F# E D# C# B A# G# F# E D# C#

Sweep 1 (ascending): $L_y > L_o > P > A$

```
G|-----1--3--4-|
D|-----1--2--4-|
A|-----1--2--4-|
E|---0--2--4-----|
```

Sweep 2 (descending): $L_o > L_y > I > M$

```
G|---6--4--3-----|
D|-----6--4--2-----|
A|-----6--4--2-----|
E|-----6--4--2-----|
```

Sweep 3 (ascending): $A > D > M > I$

```

G|-----4--6--8-|
D|-----4--6--8-|
A|-----4--6--7-|
E|---4--6--7-----|

```

Sweep 4 (descending): $D > A > P > Lo$

```

G|---9--8--6-----|
D|-----9--8--6-----|
A|-----9--7--6-----|
E|-----9--7--6-|

```

Sweep 5 (ascending): $I > Ly > Lo > P$

```

G|-----8--9-11-|
D|-----8--9-11-|
A|-----7--9-11-|
E|---7--9-11-----|

```

Sweep 6 (descending): $Ly > I > M > D$

```

G|--13-11--9-----|
D|-----13-11--9-----|
A|-----13-11--9-----|
E|-----12-11--9-|

```

F# major

Notes: F# G# A# B C# D# E#

Scale notes on the neck (frets 0-12):

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	.	G#	.	A#	B	.	C#	.	D#	.	E#	F#	.
D	.	D#	.	E#	F#	.	G#	.	A#	B	.	C#	.
A	.	A#	B	.	C#	.	D#	.	E#	F#	.	G#	.
E	.	E#	F#	.	G#	.	A#	B	.	C#	.	D#	.

Same neck, MILLPAD label of the chunk that STARTS on each note:

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	.	D	.	P	Ly	.	M	.	A	.	Lo	I	.
D	.	A	.	Lo	I	.	D	.	P	Ly	.	M	.
A	.	P	Ly	.	M	.	A	.	Lo	I	.	D	.
E	.	Lo	I	.	D	.	P	Ly	.	M	.	A	.

The full traversal, lowest playable start upward:

#	Dir	Labels (play order)	E	A	D	G	Notes (play order)
1	UP	Lo P A D	1-2-4	1-2-4	1-3-4	1-3-4	E# F# G# A# B C# D# E# F# G# A# B
2	DOWN	P Lo Ly I	6-4-2	6-4-2	6-4-3	6-4-3	C# B A# G# F# E# D# C# B A# G# F#
3	UP	D M I Ly	4-6-7	4-6-8	4-6-8	4-6-8	G# A# B C# D# E# F# G# A# B C# D#
4	DOWN	M D A P	9-7-6	9-8-6	9-8-6	10-8-6	E# D# C# B A# G# F# E# D# C# B A#
5	UP	Ly Lo P A	7-9-11	8-9-11	8-9-11	8-10-11	B C# D# E# F# G# A# B C# D# E# F#
6	DOWN	Lo Ly I M	13-11-9	13-11-9	13-11-9	13-11-10	G# F# E# D# C# B A# G# F# E# D# C#

Sweep 1 (ascending): $Lo > P > A > D$

```

G|-----1--3--4-|
D|-----1--3--4-|
A|-----1--2--4-|
E|---1--2--4-----|

```

Sweep 2 (descending): $P > Lo > Ly > I$

```

G|---6--4--3-----|
D|-----6--4--3-----|
A|-----6--4--2-----|
E|-----6--4--2-----|

```

Sweep 3 (ascending): D > M > I > Ly

```

G|-----4--6--8-----|
D|-----4--6--8-----|
A|-----4--6--8-----|
E|---4--6--7-----|

```

Sweep 4 (descending): M > D > A > P

```

G|--10--8--6-----|
D|-----9--8--6-----|
A|-----9--8--6-----|
E|-----9--7--6-----|

```

Sweep 5 (ascending): Ly > Lo > P > A

```

G|-----8-10-11-----|
D|-----8--9-11-----|
A|-----8--9-11-----|
E|---7--9-11-----|

```

Sweep 6 (descending): Lo > Ly > I > M

```

G|--13-11-10-----|
D|-----13-11--9-----|
A|-----13-11--9-----|
E|-----13-11--9-----|

```

Db major

Notes: Db Eb F Gb Ab Bb C

Scale notes on the neck (frets 0-12):

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	.	Ab	.	Bb	.	C	Db	.	Eb	.	F	Gb	.
D	.	Eb	.	F	Gb	.	Ab	.	Bb	.	C	Db	.
A	.	Bb	.	C	Db	.	Eb	.	F	Gb	.	Ab	.
E	.	F	Gb	.	Ab	.	Bb	.	C	Db	.	Eb	.

Same neck, MILLPAD label of the chunk that STARTS on each note:

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	.	M	.	A	.	Lo	I	.	D	.	P	Ly	.
D	.	D	.	P	Ly	.	M	.	A	.	Lo	I	.
A	.	A	.	Lo	I	.	D	.	P	Ly	.	M	.
E	.	P	Ly	.	M	.	A	.	Lo	I	.	D	.

The full traversal, lowest playable start upward:

#	Dir	Labels (play order)	E	A	D	G	Notes (play order)
1	UP	P A D M	1-2-4	1-3-4	1-3-4	1-3-5	F Gb Ab Bb C Db Eb F Gb Ab Bb C
2	DOWN	A P Lo Ly	6-4-2	6-4-3	6-4-3	6-5-3	Db C Bb Ab Gb F Eb Db C Bb Ab Gb
3	UP	M I Ly Lo	4-6-8	4-6-8	4-6-8	5-6-8	Ab Bb C Db Eb F Gb Ab Bb C Db Eb
4	DOWN	I M D A	9-8-6	9-8-6	10-8-6	10-8-6	F Eb Db C Bb Ab Gb F Eb Db C Bb
5	UP	Lo P A D	8-9-11	8-9-11	8-10-11	8-10-11	C Db Eb F Gb Ab Bb C Db Eb F Gb
6	DOWN	P Lo Ly I	13-11-9	13-11-9	13-11-10	13-11-10	Ab Gb F Eb Db C Bb Ab Gb F Eb Db

Sweep 1 (ascending): P > A > D > M


```
G|-----1--3--5-|
D|-----1--3--4-|
A|-----1--3--4-|
E|---1--2--4-----|
```

Sweep 2 (descending): A > P > Lo > Ly

```
G|---6--5--3-----|
D|-----6--4--3-----|
A|-----6--4--3-----|
E|-----6--4--2-----|
```

Sweep 3 (ascending): M > I > Ly > Lo

```
G|-----5--6--8-|
D|-----4--6--8-|
A|-----4--6--8-|
E|---4--6--8-----|
```

Sweep 4 (descending): I > M > D > A

```
G|--10--8--6-----|
D|-----10--8--6-----|
A|-----9--8--6-----|
E|-----9--8--6-----|
```

Sweep 5 (ascending): Lo > P > A > D

```
G|-----8-10-11-|
D|-----8-10-11-|
A|-----8--9-11-|
E|---8--9-11-----|
```

Sweep 6 (descending): P > Lo > Ly > I

```
G|--13-11-10-----|
D|-----13-11-10-----|
A|-----13-11--9-----|
E|-----13-11--9-----|
```

Ab major

Notes: Ab Bb C Db Eb F G

Scale notes on the neck (frets 0-12):

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	G	Ab	.	Bb	.	C	Db	.	Eb	.	F	.	G
D	.	Eb	.	F	.	G	Ab	.	Bb	.	C	Db	.
A	.	Bb	.	C	Db	.	Eb	.	F	.	G	Ab	.
E	.	F	.	G	Ab	.	Bb	.	C	Db	.	Eb	.

Same neck, MILLPAD label of the chunk that STARTS on each note:

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	Lo	I	.	D	.	P	Ly	.	M	.	A	.	Lo
D	.	M	.	A	.	Lo	I	.	D	.	P	Ly	.
A	.	D	.	P	Ly	.	M	.	A	.	Lo	I	.
E	.	A	.	Lo	I	.	D	.	P	Ly	.	M	.

The full traversal, lowest playable start upward:

#	Dir	Labels (play order)	E	A	D	G	Notes (play order)
1	UP	A D M I	1-3-4	1-3-4	1-3-5	1-3-5	F G Ab Bb C Db Eb F G Ab Bb C
2	DOWN	D A P Lo	6-4-3	6-4-3	6-5-3	6-5-3	Db C Bb Ab G F Eb Db C Bb Ab G
3	UP	I Ly Lo P	4-6-8	4-6-8	5-6-8	5-6-8	Ab Bb C Db Eb F G Ab Bb C Db Eb
4	DOWN	Ly I M D	9-8-6	10-8-6	10-8-6	10-8-6	F Eb Db C Bb Ab G F Eb Db C Bb
5	UP	P A D M	8-9-11	8-10-11	8-10-11	8-10-12	C Db Eb F G Ab Bb C Db Eb F G
6	DOWN	A P Lo Ly	13-11-9	13-11-10	13-11-10	13-12-10	Ab G F Eb Db C Bb Ab G F Eb Db

Sweep 1 (ascending): A > D > M > I

G -----1--3--5-
D -----1--3--5-----
A -----1--3--4-----
E ---1--3--4-----

Sweep 2 (descending): D > A > P > Lo

G ---6--5--3-----
D -----6--5--3-----
A -----6--4--3-----
E -----6--4--3-

Sweep 3 (ascending): I > Ly > Lo > P

G -----5--6--8-
D -----5--6--8-----
A -----4--6--8-----
E ---4--6--8-----

Sweep 4 (descending): Ly > I > M > D

G --10--8--6-----
D -----10--8--6-----
A -----10--8--6-----
E -----9--8--6-

Sweep 5 (ascending): P > A > D > M

G -----8-10-12-
D -----8-10-11-----
A -----8-10-11-----
E ---8--9-11-----

Sweep 6 (descending): A > P > Lo > Ly

G --13-12-10-----
D -----13-11-10-----
A -----13-11-10-----
E -----13-11--9-

Eb major

Notes: Eb F G Ab Bb C D

Scale notes on the neck (frets 0-12):

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	G	Ab	.	Bb	.	C	.	D	Eb	.	F	.	G
D	D	Eb	.	F	.	G	Ab	.	Bb	.	C	.	D
A	.	Bb	.	C	.	D	Eb	.	F	.	G	Ab	.
E	.	F	.	G	Ab	.	Bb	.	C	.	D	Eb	.

Same neck, MILLPAD label of the chunk that STARTS on each note:

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	P	Ly	.	M	.	A	.	Lo	I	.	D	.	P
D	Lo	I	.	D	.	P	Ly	.	M	.	A	.	Lo
A	.	M	.	A	.	Lo	I	.	D	.	P	Ly	.
E	.	D	.	P	Ly	.	M	.	A	.	Lo	I	.

The full traversal, lowest playable start upward:

#	Dir	Labels (play order)	E	A	D	G	Notes (play order)
1	UP	D M I Ly	1-3-4	1-3-5	1-3-5	1-3-5	F G Ab Bb C D Eb F G Ab Bb C
2	DOWN	M D A P	6-4-3	6-5-3	6-5-3	7-5-3	D C Bb Ab G F Eb D C Bb Ab G
3	UP	Ly Lo P A	4-6-8	5-6-8	5-6-8	5-7-8	Ab Bb C D Eb F G Ab Bb C D Eb
4	DOWN	Lo Ly I M	10-8-6	10-8-6	10-8-6	10-8-7	F Eb D C Bb Ab G F Eb D C Bb
5	UP	A D M I	8-10-11	8-10-11	8-10-12	8-10-12	C D Eb F G Ab Bb C D Eb F G
6	DOWN	D A P Lo	13-11-10	13-11-10	13-12-10	13-12-10	Ab G F Eb D C Bb Ab G F Eb D

Sweep 1 (ascending): D > M > I > Ly

G	-----1--3--5-
D	-----1--3--5-
A	-----1--3--5-
E	---1--3--4-----

Sweep 2 (descending): M > D > A > P

G	---7--5--3-----
D	-----6--5--3-
A	-----6--5--3-
E	-----6--4--3-

Sweep 3 (ascending): Ly > Lo > P > A

G	-----5--7--8-
D	-----5--6--8-
A	-----5--6--8-
E	---4--6--8-----

Sweep 4 (descending): Lo > Ly > I > M

G	--10--8--7-----
D	-----10--8--6-
A	-----10--8--6-
E	-----10--8--6-

Sweep 5 (ascending): A > D > M > I

G	-----8-10-12-
D	-----8-10-12-
A	-----8-10-11-
E	---8-10-11-----

Sweep 6 (descending): D > A > P > Lo

G	--13-12-10-----
D	-----13-12-10-
A	-----13-11-10-
E	-----13-11-10-

Bb major

Notes: Bb C D Eb F G A

Scale notes on the neck (frets 0-12):

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	G	.	A	Bb	.	C	.	D	Eb	.	F	.	G
D	D	Eb	.	F	.	G	.	A	Bb	.	C	.	D
A	A	Bb	.	C	.	D	Eb	.	F	.	G	.	A
E	.	F	.	G	.	A	Bb	.	C	.	D	Eb	.

Same neck, MILLPAD label of the chunk that STARTS on each note:

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	A	.	Lo	I	.	D	.	P	Ly	.	M	.	A
D	P	Ly	.	M	.	A	.	Lo	I	.	D	.	P
A	Lo	I	.	D	.	P	Ly	.	M	.	A	.	Lo
E	.	M	.	A	.	Lo	I	.	D	.	P	Ly	.

The full traversal, lowest playable start upward:

#	Dir	Labels (play order)	E	A	D	G	Notes (play order)
1	UP	M I Ly Lo	1-3-5	1-3-5	1-3-5	2-3-5	F G A Bb C D Eb F G A Bb C
2	DOWN	I M D A	6-5-3	6-5-3	7-5-3	7-5-3	D C Bb A G F Eb D C Bb A G
3	UP	Lo P A D	5-6-8	5-6-8	5-7-8	5-7-8	A Bb C D Eb F G A Bb C D Eb
4	DOWN	P Lo Ly I	10-8-6	10-8-6	10-8-7	10-8-7	F Eb D C Bb A G F Eb D C Bb
5	UP	D M I Ly	8-10-11	8-10-12	8-10-12	8-10-12	C D Eb F G A Bb C D Eb F G
6	DOWN	M D A P	13-11-10	13-12-10	13-12-10	14-12-10	A G F Eb D C Bb A G F Eb D

Sweep 1 (ascending): M > I > Ly > Lo

G	-----2--3--5-
D	-----1--3--5-----
A	-----1--3--5-----
E	---1--3--5-----

Sweep 2 (descending): I > M > D > A

G	---7--5--3-----
D	-----7--5--3-----
A	-----6--5--3-----
E	-----6--5--3-

Sweep 3 (ascending): Lo > P > A > D

G	-----5--7--8-
D	-----5--7--8-----
A	-----5--6--8-----
E	---5--6--8-----

Sweep 4 (descending): P > Lo > Ly > I

G	--10--8--7-----
D	-----10--8--7-----
A	-----10--8--6-----
E	-----10--8--6-

Sweep 5 (ascending): D > M > I > Ly

G	-----8-10-12-
D	-----8-10-12-----
A	-----8-10-12-----
E	---8-10-11-----

Sweep 6 (descending): M > D > A > P

G	--14-12-10-----
D	-----13-12-10-----
A	-----13-12-10-----
E	-----13-11-10-

F major

Notes: F G A Bb C D E

Scale notes on the neck (frets 0-12):

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	G	.	A	Bb	.	C	.	D	.	E	F	.	G
D	D	.	E	F	.	G	.	A	Bb	.	C	.	D
A	A	Bb	.	C	.	D	.	E	F	.	G	.	A
E	E	F	.	G	.	A	Bb	.	C	.	D	.	E

Same neck, MILLPAD label of the chunk that STARTS on each note:

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	D	.	P	Ly	.	M	.	A	.	Lo	I	.	D
D	A	.	Lo	I	.	D	.	P	Ly	.	M	.	A
A	P	Ly	.	M	.	A	.	Lo	I	.	D	.	P
E	Lo	I	.	D	.	P	Ly	.	M	.	A	.	Lo

The full traversal, lowest playable start upward:

#	Dir	Labels (play order)	E	A	D	G	Notes (play order)
1	UP	Lo P A D	0-1-3	0-1-3	0-2-3	0-2-3	E F G A Bb C D E F G A Bb
2	DOWN	P Lo Ly I	5-3-1	5-3-1	5-3-2	5-3-2	C Bb A G F E D C Bb A G F
3	UP	D M I Ly	3-5-6	3-5-7	3-5-7	3-5-7	G A Bb C D E F G A Bb C D
4	DOWN	M D A P	8-6-5	8-7-5	8-7-5	9-7-5	E D C Bb A G F E D C Bb A
5	UP	Ly Lo P A	6-8-10	7-8-10	7-8-10	7-9-10	Bb C D E F G A Bb C D E F
6	DOWN	Lo Ly I M	12-10-8	12-10-8	12-10-8	12-10-9	G F E D C Bb A G F E D C

Sweep 1 (ascending): Lo > P > A > D

```
G|-----0--2--3-|
D|-----0--2--3-----|
A|-----0--1--3-----|
E|---0--1--3-----|
```

Sweep 2 (descending): P > Lo > Ly > I

```
G|---5--3--2-----|
D|-----5--3--2-----|
A|-----5--3--1-----|
E|-----5--3--1-|
```

Sweep 3 (ascending): D > M > I > Ly

```
G|-----3--5--7-|
D|-----3--5--7-----|
A|-----3--5--7-----|
E|---3--5--6-----|
```

Sweep 4 (descending): M > D > A > P

```
G|---9--7--5-----|
D|-----8--7--5-----|
A|-----8--7--5-----|
E|-----8--6--5-|
```

Sweep 5 (ascending): Ly > Lo > P > A

```
G|-----7--9-10-|
D|-----7--8-10-----|
A|-----7--8-10-----|
E|---6--8-10-----|
```

Sweep 6 (descending): Lo > Ly > I > M

```

G| --12-10--9-----|
D| -----12-10--8-----|
A| -----12-10--8-----|
E| -----12-10--8-----|

```

2.7 The seven starting rows in the seven practice keys

The single densest reference in this manual. For each key, every one of the seven starting rows at its lowest playable position: the four labels, the twelve frets and where the shift falls.

Generate a row before you look it up. Using this table as a first resort will cost you the entire point of the month.

C major — C D E F G A B

Start	Labels	E	A	D	G	Shift
M	M I Ly Lo	3-5-7	3-5-7	3-5-7	4-5-7	G str
I	I Ly Lo P	8-10-12	8-10-12	9-10-12	9-10-12	D str
Ly	Ly Lo P A	1-3-5	2-3-5	2-3-5	2-4-5	A str
Lo	Lo P A D	7-8-10	7-8-10	7-9-10	7-9-10	none
P	P A D M	0-1-3	0-2-3	0-2-3	0-2-4	none
A	A D M I	5-7-8	5-7-8	5-7-9	5-7-9	none
D	D M I Ly	10-12-13	10-12-14	10-12-14	10-12-14	none

G major — G A B C D E F#

Start	Labels	E	A	D	G	Shift
M	M I Ly Lo	10-12-14	10-12-14	10-12-14	11-12-14	G str
I	I Ly Lo P	3-5-7	3-5-7	4-5-7	4-5-7	D str
Ly	Ly Lo P A	8-10-12	9-10-12	9-10-12	9-11-12	A str
Lo	Lo P A D	2-3-5	2-3-5	2-4-5	2-4-5	none
P	P A D M	7-8-10	7-9-10	7-9-10	7-9-11	none
A	A D M I	0-2-3	0-2-3	0-2-4	0-2-4	none
D	D M I Ly	5-7-8	5-7-9	5-7-9	5-7-9	none

F major — F G A Bb C D E

Start	Labels	E	A	D	G	Shift
M	M I Ly Lo	8-10-12	8-10-12	8-10-12	9-10-12	G str
I	I Ly Lo P	1-3-5	1-3-5	2-3-5	2-3-5	D str
Ly	Ly Lo P A	6-8-10	7-8-10	7-8-10	7-9-10	A str
Lo	Lo P A D	0-1-3	0-1-3	0-2-3	0-2-3	none
P	P A D M	5-6-8	5-7-8	5-7-8	5-7-9	none
A	A D M I	10-12-13	10-12-13	10-12-14	10-12-14	none
D	D M I Ly	3-5-6	3-5-7	3-5-7	3-5-7	none

D major — D E F# G A B C#

Start	Labels	E	A	D	G	Shift
M	M I Ly Lo	5-7-9	5-7-9	5-7-9	6-7-9	G str
I	I Ly Lo P	10-12-14	10-12-14	11-12-14	11-12-14	D str
Ly	Ly Lo P A	3-5-7	4-5-7	4-5-7	4-6-7	A str
Lo	Lo P A D	9-10-12	9-10-12	9-11-12	9-11-12	none
P	P A D M	2-3-5	2-4-5	2-4-5	2-4-6	none
A	A D M I	7-9-10	7-9-10	7-9-11	7-9-11	none
D	D M I Ly	0-2-3	0-2-4	0-2-4	0-2-4	none

Bb major — Bb C D Eb F G A

Start	Labels	E	A	D	G	Shift
M	M I Ly Lo	1-3-5	1-3-5	1-3-5	2-3-5	G str
I	I Ly Lo P	6-8-10	6-8-10	7-8-10	7-8-10	D str
Ly	Ly Lo P A	11-13-15	12-13-15	12-13-15	12-14-15	A str
Lo	Lo P A D	5-6-8	5-6-8	5-7-8	5-7-8	none
P	P A D M	10-11-13	10-12-13	10-12-13	10-12-14	none
A	A D M I	3-5-6	3-5-6	3-5-7	3-5-7	none
D	D M I Ly	8-10-11	8-10-12	8-10-12	8-10-12	none

A major — A B C# D E F# G#

Start	Labels	E	A	D	G	Shift
M	M I Ly Lo	0-2-4	0-2-4	0-2-4	1-2-4	G str
I	I Ly Lo P	5-7-9	5-7-9	6-7-9	6-7-9	D str
Ly	Ly Lo P A	10-12-14	11-12-14	11-12-14	11-13-14	A str
Lo	Lo P A D	4-5-7	4-5-7	4-6-7	4-6-7	none
P	P A D M	9-10-12	9-11-12	9-11-12	9-11-13	none
A	A D M I	2-4-5	2-4-5	2-4-6	2-4-6	none
D	D M I Ly	7-9-10	7-9-11	7-9-11	7-9-11	none

Eb major — Eb F G Ab Bb C D

Start	Labels	E	A	D	G	Shift
M	M I Ly Lo	6-8-10	6-8-10	6-8-10	7-8-10	G str
I	I Ly Lo P	11-13-15	11-13-15	12-13-15	12-13-15	D str
Ly	Ly Lo P A	4-6-8	5-6-8	5-6-8	5-7-8	A str
Lo	Lo P A D	10-11-13	10-11-13	10-12-13	10-12-13	none
P	P A D M	3-4-6	3-5-6	3-5-6	3-5-7	none
A	A D M I	8-10-11	8-10-11	8-10-12	8-10-12	none
D	D M I Ly	1-3-4	1-3-5	1-3-5	1-3-5	none

PART THREE — THE THIRTY-DAY PROGRAMME

How this programme is built

Sixty minutes a day, thirty days. Every session has the same seven-block skeleton so you never have to decide what to do; you only have to do it.

1. Hands and tuning	3 min	never skipped
2. Fretboard naming	8 min	the actual bottleneck
3. Shape work	8 min	mechanical, no thinking
4. MILLPAD verbal	5 min	away from the bass
5. Main material	22 min	the day's new work
6. Exploration passage	10 min	from Part Four
7. Repair and log	4 min	non-negotiable

The proportions shift as you progress — later weeks give more to block 5 and less to blocks 2 and 3 — but the skeleton never changes.

Three rules that decide whether this works

1. **Speak while you play.** Every drill in this programme has something to say out loud: a note name, a label, a direction. Silent practice is finger practice; spoken practice is navigation practice. You are training the second one.
2. **Advance on three clean repetitions, not on one.** “Clean” means correct notes, even timing, no ringing strings, relaxed hands. Two out of four is not a pass.
3. **Never restart from the beginning after a mistake.** Restart from the nearest checkpoint. Restarting from the top over-trains the opening and under-trains the failure point, which is the single most common reason people plateau on this exercise.

The tempo ladder

Use it everywhere. When a new element is added, drop two rungs and climb again.

Rung	Metronome	Subdivision	Advance when
1	40	one note per click	3 clean
2	48	one note per click	3 clean
3	56	one note per click	3 clean
4	63	one note per click	3 clean
5	40	two notes per click	3 clean
6	48	two notes per click	3 clean
7	56	two notes per click	3 clean
8	63	two notes per click	3 clean
9	72	two notes per click	3 clean
10	80	two notes per click	final target

Rung 8 at day 30 is a genuine, musically useful standard. Chasing rung 10 by day 12 will cost you the whole month.

Phase map

Days	Phase	What changes
1-5	Foundation	Note names and the three shapes, separately
6-8	Linking	MILLPAD cycle bolted onto the shapes
9-13	Vertical sweeps	All seven starting rows, C major, no turnarounds
14-17	Turnarounds and full cycle	C major, open position to fret 12
18-23	Transposition	G, F, D, B-flat, A, E-flat
24-26	Free navigation	Any key, any fret, random starts
27-30	Musical integration	Modes as sound, then final assessment

PHASE 1 — FOUNDATION (Days 1-5)

The reason people fail The Beast is almost never the fingering. It is that they cannot name the note under their finger fast enough to keep the scale running. Five days on this now saves you three weeks later.

Day 1 — Foundation

Today's single objective: Name any note on the E string in under two seconds, and play the WS-WS shape without looking.

The E string is the lowest string, where every sweep begins.

Natural notes on the E string: E=0, F=1, G=3, A=5, B=7, C=8, D=10, E=12

Anchors to burn in first: fret 0, fret 5, fret 7, fret 12. On the E string those are 0=E, 5=A, 7=B, 12=E.

Block	Min	Content	Standard to hit
Hands	3	Open-string alternation, index/middle, four cycles. Loose shoulders.	No forearm tension
Naming	12	Coordinate-to-note: 20 random frets on the E string, say before you play. Then note-to-coordinate: 10 note names, find them.	18 of 20 correct, under 2 sec each
Shape	12	Passage E1 on the E string, tempo rungs 1-3.	3 clean at rung 3
Octaves	8	From each natural note on the E string, play its octave (two strings up, two frets across) where available.	12 pairs, no hunting
MILLPAD	5	Say M-I-Ly-Lo-P-A-D forwards ten times, then backwards ten times. Away from the bass.	No hesitation
Silent map	5	Eyes shut. Picture the E string. Walk fret 0 to 12 naming every note.	Two complete passes
Log	5	Write the naming score, the tempo rung reached, and one thing that felt bad.	Written down, not remembered
Total	50		

Exit test for Day 1: twenty random coordinates on the E string, eighteen correct, each under two seconds. If you fail, repeat this day tomorrow. Repeating a foundation day is not falling behind; arriving at Day 14 unable to name notes is.

Day 2 — Foundation

Today's single objective: Name any note on the A string in under two seconds, and play the 1-2-4 shape without looking.

The A string is the string you cross to first, and therefore the one you lose.

Natural notes on the A string: A=0, B=2, C=3, D=5, E=7, F=8, G=10, A=12

Anchors to burn in first: fret 0, fret 5, fret 7, fret 12. On the A string those are 0=A, 5=D, 7=E, 12=A.

Block	Min	Content	Standard to hit
Hands	3	Open-string alternation, index/middle, four cycles. Loose shoulders.	No forearm tension
Naming	12	Coordinate-to-note: 20 random frets on the A string, say before you play. Then note-to-coordinate: 10 note names, find them.	18 of 20 correct, under 2 sec each
Shape	12	Passage E2 on the A string, tempo rungs 1-3.	3 clean at rung 3
Octaves	8	From each natural note on the A string, play its octave (two strings up, two frets across) where available.	12 pairs, no hunting
MILLPAD	5	Say M-I-Ly-Lo-P-A-D forwards ten times, then backwards ten times. Away from the bass.	No hesitation
Silent map	5	Eyes shut. Picture the A string. Walk fret 0 to 12 naming every note.	Two complete passes
Log	5	Write the naming score, the tempo rung reached, and one thing that felt bad.	Written down, not remembered
Total	50		

Exit test for Day 2: twenty random coordinates on the A string, eighteen correct, each under two seconds. If you fail, repeat this day tomorrow. Repeating a foundation day is not falling behind; arriving at Day 14 unable to name notes is.

Day 3 — Foundation

Today's single objective: Name any note on the D string in under two seconds, and play the 1-3-4 shape without looking.

The D string is the middle of the sweep, where attention typically drops.

Natural notes on the D string: D=0, E=2, F=3, G=5, A=7, B=9, C=10, D=12

Anchors to burn in first: fret 0, fret 5, fret 7, fret 12. On the D string those are 0=D, 5=G, 7=A, 12=D.

Block	Min	Content	Standard to hit
Hands	3	Open-string alternation, index/middle, four cycles. Loose shoulders.	No forearm tension
Naming	12	Coordinate-to-note: 20 random frets on the D string, say before you play. Then note-to-coordinate: 10 note names, find them.	18 of 20 correct, under 2 sec each
Shape	12	Passage E3 on the D string, tempo rungs 1-3.	3 clean at rung 3
Octaves	8	From each natural note on the D string, play its octave (two strings up, two frets across) where available.	12 pairs, no hunting
MILLPAD	5	Say M-I-Ly-Lo-P-A-D forwards ten times, then backwards ten times. Away from the bass.	No hesitation
Silent map	5	Eyes shut. Picture the D string. Walk fret 0 to 12 naming every note.	Two complete passes
Log	5	Write the naming score, the tempo rung reached, and one thing that felt bad.	Written down, not remembered
Total	50		

Exit test for Day 3: twenty random coordinates on the D string, eighteen correct, each under two seconds. If you fail, repeat this day tomorrow. Repeating a foundation day is not falling behind; arriving at Day 14 unable to name notes is.

Day 4 — Foundation

Today's single objective: Name any note on the G string in under two seconds, and play the WS-WS shape without looking.

The G string is the turnaround string — the highest-stakes real estate on the neck.

Natural notes on the G string: G=0, A=2, B=4, C=5, D=7, E=9, F=10, G=12

Anchors to burn in first: fret 0, fret 5, fret 7, fret 12. On the G string those are 0=G, 5=C, 7=D, 12=G.

Block	Min	Content	Standard to hit
Hands	3	Open-string alternation, index/middle, four cycles. Loose shoulders.	No forearm tension
Naming	12	Coordinate-to-note: 20 random frets on the G string, say before you play. Then note-to-coordinate: 10 note names, find them.	18 of 20 correct, under 2 sec each
Shape	12	Passage E1 on the G string, tempo rungs 1-3.	3 clean at rung 3
Octaves	8	From each natural note on the G string, play its octave (two strings up, two frets across) where available.	12 pairs, no hunting
MILLPAD	5	Say M-I-Ly-Lo-P-A-D forwards ten times, then backwards ten times. Away from the bass.	No hesitation
Silent map	5	Eyes shut. Picture the G string. Walk fret 0 to 12 naming every note.	Two complete passes
Log	5	Write the naming score, the tempo rung reached, and one thing that felt bad.	Written down, not remembered
Total	50		

Exit test for Day 4: twenty random coordinates on the G string, eighteen correct, each under two seconds. If you fail, repeat this day tomorrow. Repeating a foundation day is not falling behind; arriving at Day 14 unable to name notes is.

Day 5 — Foundation

Today's single objective: Name any note anywhere on the neck, and switch between all three shapes on command.

The full chromatic neck, frets 0-12:

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	G	G#/Ab	A	A#/Bb	B	C	C#/Db	D	D#/Eb	E	F	F#/Gb	G
D	D	D#/Eb	E	F	F#/Gb	G	G#/Ab	A	A#/Bb	B	C	C#/Db	D
A	A	A#/Bb	B	C	C#/Db	D	D#/Eb	E	F	F#/Gb	G	G#/Ab	A
E	E	F	F#/Gb	G	G#/Ab	A	A#/Bb	B	C	C#/Db	D	D#/Eb	E

The same neck showing natural notes only. This is the skeleton — everything else is a natural note plus or minus one fret.

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	G	.	A	.	B	C	.	D	.	E	F	.	G
D	D	.	E	F	.	G	.	A	.	B	C	.	D
A	A	.	B	C	.	D	.	E	F	.	G	.	A
E	E	F	.	G	.	A	.	B	C	.	D	.	E

Two facts collapse this whole table into something memorable:

- **B to C and E to F are one fret apart.** Every other pair of adjacent letters is two frets.
- **Fret 5 on any string equals the next open string up** (A, D, G, C). **Fret 12 is the open string an octave higher.**

Block	Min	Content	Standard to hit
Hands	3	Open-string alternation and relaxed shifts between fret 1 and fret 9.	No thumb anchoring
Naming	12	Random string AND random fret, 30 prompts. Then: pick one note name, find all its locations 0-12.	27 of 30, under 2 sec
Shape switching	15	Random shape called every 4 bars: WS-WS, 1-2-4, 1-3-4. Play it from fret 5 on a random string. Passages E1-E4.	20 switches, no wrong shape
Fifth-fret web	8	Play fret 5 on E, then open A — same pitch. Repeat for A/D and D/G. Then fret 7 landmarks: B, E, A, D.	Instant recall
MILLPAD	5	Recite forwards and backwards. Then: given a label, say its shape. 20 prompts.	20 of 20
Silent map	7	Eyes shut, name every note at fret 3, then fret 8, then fret 10, across all four strings.	No hesitation
Log	5	Record scores. Note which string is slowest — that string gets extra time tomorrow.	Written
Total	55		

Phase 1 exit standard (must pass before Day 6):

- 30 random coordinates, 27 correct, under two seconds each.
- All three shapes played from a called fret on a called string, 20 out of 20.
- MILLPAD spoken forwards and backwards without pause.
- Every location of the note C between frets 0 and 12, named without the chart.

PHASE 2 — LINKING (Days 6-8)

The shapes exist. The note names exist. Now bolt the label track on. Three days, almost all of it verbal. You will play less this week than any other, and it will pay for itself twice over in Phase 3.

Day 6 — Linking

Today's single objective: Convert MILLPAD from a word you can say into a lookup table you can query in both directions.

The master grid. Learn it as three-two-two, not as seven separate facts.

MILLPAD	Mode	Degree in parent key	First two steps	Shape	Fret offsets	Fingers
M	Mixolydian	5	W-W	WS-WS	0, +2, +4	1-3-4 high / 1-2-4 low + shift
I	Ionian	1	W-W	WS-WS	0, +2, +4	1-3-4 high / 1-2-4 low + shift
Ly	Lydian	4	W-W	WS-WS	0, +2, +4	1-3-4 high / 1-2-4 low + shift
Lo	Locrian	7	H-W	1-2-4	0, +1, +3	1-2-4
P	Phrygian	3	H-W	1-2-4	0, +1, +3	1-2-4
A	Aeolian	6	W-H	1-3-4	0, +2, +3	1-3-4
D	Dorian	2	W-H	1-3-4	0, +2, +3	1-3-4

```
M   I   Ly | Lo  P  | A   D
W-W W-W W-W | H-W H-W| W-H W-H
<--- three ---> <- two -> <- two ->
```

Three query directions. You need all three, and most people only ever train the first.

1. **Label to shape.** “Locrian” -> 1-2-4, H-W, x/x+1/x+3.
2. **Shape to labels.** “W-H” -> Aeolian or Dorian.
3. **Label to next label.** “Lydian” -> next is Locrian, **and the hand steps up one fret.**

Block	Min	Content	Standard to hit
Hands	3	Warm-up, any comfortable shape at fret 7.	Relaxed
Naming	6	Maintenance: 15 random coordinates.	14 of 15
Label to shape	12	Random label called, you state the shape and play it from fret 5 on a random string. 30 prompts.	28 of 30
Shape to label	10	Someone plays or you read a shape; you name both possible labels. 20 prompts.	18 of 20
Label to next	12	Random label, you say the next one forwards, then the next one backwards. Flag the Ly-to-Lo shift every time it comes up.	30 prompts, no misses
Cycle drill	12	Recite MILLPAD starting from each of the seven labels in turn, forwards then backwards.	14 recitations clean
Log	5	Which label do you hesitate on? That one gets doubled tomorrow.	Written
Total	60		

Exit test: thirty random prompts mixing all three query directions, twenty-eight correct.

Day 7 — Linking — checkpoint

Today's single objective: Attach the labels to actual scale degrees in C major, on the neck.

C major on the neck, as note names:

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	G	.	A	.	B	C	.	D	.	E	F	.	G
D	D	.	E	F	.	G	.	A	.	B	C	.	D
A	A	.	B	C	.	D	.	E	F	.	G	.	A
E	E	F	.	G	.	A	.	B	C	.	D	.	E

The identical neck, as MILLPAD labels. Each cell says: *if a three-note chunk starts here, this is its label and therefore its shape.*

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	M	.	A	.	Lo	I	.	D	.	P	Ly	.	M
D	D	.	P	Ly	.	M	.	A	.	Lo	I	.	D
A	A	.	Lo	I	.	D	.	P	Ly	.	M	.	A
E	P	Ly	.	M	.	A	.	Lo	I	.	D	.	P

And as scale degrees, which is the layer that actually transposes:

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	5	.	6	.	7	1	.	2	.	3	4	.	5
D	2	.	3	4	.	5	.	6	.	7	1	.	2
A	6	.	7	1	.	2	.	3	4	.	5	.	6
E	3	4	.	5	.	6	.	7	1	.	2	.	3

Study the three maps together until you can generate the second and third from the first. The degree map is the one that survives a key change — the note names change, the degree map does not move relative to the tonic.

Block	Min	Content	Standard to hit
Hands	3	Warm-up.	Relaxed
Naming	6	Maintenance: 15 coordinates.	14 of 15
Degree drill	12	Random coordinate in C major: say the note, then its degree, then its label, then its shape. 25 prompts.	22 of 25
Shape placement	12	Called a coordinate, play the correct three-note chunk that STARTS there. 20 prompts.	18 of 20
Checkpoint test	15	Phase 1 exit standard re-run in full, plus Day 6 exit test.	Both passed
MILLPAD	7	Seven-start recitation, forwards and backwards.	Clean
Log	5	Record both test scores. If either failed, tomorrow is a repeat of the failed day.	Written
Total	60		

Day 8 — Linking

Today's single objective: Play any single chunk in C major from any starting note, correctly labelled, without hunting.

This is the last day before the sweeps begin. Today you play **one chunk at a time**, chosen at random, anywhere on the neck.

The seven chunks of C major, at their lowest position on each string. Note that the same chunk has the same shape everywhere it appears — only the fret column moves.

Label	Mode	Notes	Shape	On E	On A	On D	On G
M	Mixolydian	G A B	WS-WS	3-5-7	10-12-14	5-7-9	0-2-4
I	Ionian	C D E	WS-WS	8-10-12	3-5-7	10-12-14	5-7-9
Ly	Lydian	F G A	WS-WS	1-3-5	8-10-12	3-5-7	10-12-14
Lo	Locrian	B C D	1-2-4	7-8-10	2-3-5	9-10-12	4-5-7
P	Phrygian	E F G	1-2-4	0-1-3	7-8-10	2-3-5	9-10-12
A	Aeolian	A B C	1-3-4	5-7-8	0-2-3	7-9-10	2-4-5
D	Dorian	D E F	1-3-4	10-12-13	5-7-8	0-2-3	7-9-10

Block	Min	Content	Standard to hit
Hands	3	Warm-up.	Relaxed
Naming	5	Maintenance: 12 coordinates.	Clean
Random chunk	20	Random label AND random string called. Play that chunk in C major at its lowest available position, then its next position up the neck. 30 prompts.	27 of 30 inside 3 sec
Chunk chaining	12	Play a chunk, then say the next label forwards, then play it on the next string up. Two-chunk chains only. 15 chains.	No wrong shape
Passages	12	E5 (plain sweep) read slowly with labels spoken. Do not attempt to memorise it.	Labels spoken correctly
MILLPAD	3	Seven-start recitation.	Clean
Log	5	Record. Note any chunk that is consistently slow.	Written
Total	60		

Phase 2 exit standard: given any label and any string, play the correct C-major chunk in under three seconds, twenty-seven times out of thirty. Given any label, name the next and previous label and state whether a fret shift occurs.

PHASE 3 — VERTICAL SWEEPS (Days 9-13)

Four strings, twelve notes, no turnarounds. You are learning to cross three string boundaries while the label track keeps running. Turnarounds are deliberately excluded until Phase 4 — adding them now doubles the failure modes at exactly the wrong moment.

There are exactly **seven** starting rows, one for each MILLPAD label. Learn them as seven, not as an infinite number of positions.

E-string label	E string	A string	D string	G string	Shapes E-A-D-G	Fret columns
M	M	I	Ly	Lo	W-W W-W W-W H-W	x x x x+1
I	I	Ly	Lo	P	W-W W-W H-W H-W	x x x+1 x+1
Ly	Ly	Lo	P	A	W-W H-W H-W W-H	x x+1 x+1 x+1
Lo	Lo	P	A	D	H-W H-W W-H W-H	x x x x
P	P	A	D	M	H-W W-H W-H W-W	x x x x
A	A	D	M	I	W-H W-H W-W W-W	x x x x
D	D	M	I	Ly	W-H W-W W-W W-W	x x x x

Read the fret-column column carefully. **Four of the seven rows never shift at all.** The rows starting on Lo, P, A and D sit in one fret column for the entire sweep, because the Ly-to-Lo boundary falls outside their four strings. Those four rows are the easy ones; do them first.

Day 9 — Vertical sweeps

Today's single objective: Play the Lo/P starting row cleanly in C major, labels spoken, at rung 3.

The two no-shift rows starting on the half-step-first shape.

Row starting on Lo (Locrian) — C major, lowest position

Labels: **Lo > P > A > D**. Starting frets E-A-D-G: 7, 7, 7, 7. There is no fret shift anywhere in this row.

```
G|-----7--9-10-|
D|-----7--9-10-----|
A|-----7--8-10-----|
E|---7--8-10-----|
```

Notes: B C D | E F G | A B C | D E F

Row starting on P (Phrygian) — C major, lowest position

Labels: **P > A > D > M**. Starting frets E-A-D-G: 0, 0, 0, 0. There is no fret shift anywhere in this row.

```
G|-----0--2--4-|
D|-----0--2--3-----|
A|-----0--2--3-----|
E|---0--1--3-----|
```

Notes: E F G | A B C | D E F | G A B

The same row an octave up — identical labels, identical shapes, new column:

```
G|-----12-14-16-|
D|-----12-14-15-----|
A|-----12-14-15-----|
E|---12-13-15-----|
```

The six-pass protocol. Apply it to every row, every time. Never skip passes one to three just because you feel ready.

1. Say the four labels out loud before touching the bass.
2. Say the four starting frets.
3. Air-finger the twelve notes without plucking.
4. Play one note per click, saying the note names.
5. Play again, saying only the chunk labels.
6. Play again in silence, hearing the labels internally.

Block	Min	Content	Standard to hit
Hands	3	Warm-up plus one pass of each shape at fret 7.	Relaxed
Naming	5	Maintenance: 12 random coordinates.	11 of 12
Shape	5	Passages E1-E3, rung 4, one pass each.	Clean
Main material	24	Six-pass protocol on today's row(s). Then tempo ladder rungs 1 to 3.	3 clean at rung 3
Exploration	13	E6 pivot chunks — applied to today's row.	Labels still spoken correctly
Repair	5	Isolate the worst crossing. Play only the last note of the old string and the first note of the new one, eight times.	No hesitation
Log	5	Rung reached, error codes, worst crossing.	Written
Total	60		

Day 10 — Vertical sweeps

Today's single objective: Play the A/D starting row cleanly in C major, labels spoken, at rung 3.

The two no-shift rows starting on the half-step-last shape.

Row starting on A (Aeolian) — C major, lowest position

Labels: **A > D > M > I**. Starting frets E-A-D-G: 5, 5, 5, 5. There is no fret shift anywhere in this row.

```
G|-----5--7--9-|
D|-----5--7--9-|
A|-----5--7--8-|
E|---5--7--8-----|
```

Notes: A B C | D E F | G A B | C D E

Row starting on D (Dorian) — C major, lowest position

Labels: **D > M > I > Ly**. Starting frets E-A-D-G: 10, 10, 10, 10. There is no fret shift anywhere in this row.

```
G|-----10-12-14-|
D|-----10-12-14-|
A|-----10-12-14-|
E|--10-12-13-----|
```

Notes: D E F | G A B | C D E | F G A

The six-pass protocol. Apply it to every row, every time. Never skip passes one to three just because you feel ready.

1. Say the four labels out loud before touching the bass.
2. Say the four starting frets.
3. Air-finger the twelve notes without plucking.
4. Play one note per click, saying the note names.
5. Play again, saying only the chunk labels.
6. Play again in silence, hearing the labels internally.

Block	Min	Content	Standard to hit
Hands	3	Warm-up plus one pass of each shape at fret 7.	Relaxed
Naming	5	Maintenance: 12 random coordinates.	11 of 12
Shape	5	Passages E1-E3, rung 4, one pass each.	Clean
Main material	24	Six-pass protocol on today's row(s). Then tempo ladder rungs 1 to 3.	3 clean at rung 3
Exploration	13	E7 broken chunks — applied to today's row.	Labels still spoken correctly
Repair	5	Isolate the worst crossing. Play only the last note of the old string and the first note of the new one, eight times.	No hesitation
Log	5	Rung reached, error codes, worst crossing.	Written
Total	60		

Day 11 — Vertical sweeps

Today's single objective: Play the M starting row cleanly in C major, labels spoken, at rung 3.

The first row that contains the shift, at the very last crossing.

Row starting on M (Mixolydian) — C major, lowest position

Labels: **M** > **I** > **Ly** > **Lo**. Starting frets E-A-D-G: 3, 3, 3, 4. There is fret shift of +1 when crossing to the G string (Ly to Lo).

```
G|-----4--5--7-|
D|-----3--5--7-----|
A|-----3--5--7-----|
E|---3--5--7-----|
```

Notes: G A B | C D E | F G A | B C D

The six-pass protocol. Apply it to every row, every time. Never skip passes one to three just because you feel ready.

1. Say the four labels out loud before touching the bass.
2. Say the four starting frets.
3. Air-finger the twelve notes without plucking.
4. Play one note per click, saying the note names.
5. Play again, saying only the chunk labels.
6. Play again in silence, hearing the labels internally.

Block	Min	Content	Standard to hit
Hands	3	Warm-up plus one pass of each shape at fret 7.	Relaxed
Naming	5	Maintenance: 12 random coordinates.	11 of 12
Shape	5	Passages E1-E3, rung 4, one pass each.	Clean
Main material	24	Six-pass protocol on today's row(s). Then tempo ladder rungs 1 to 3.	3 clean at rung 3
Exploration	13	E8 retrograde — applied to today's row.	Labels still spoken correctly
Repair	5	Isolate the worst crossing. Play only the last note of the old string and the first note of the new one, eight times.	No hesitation
Log	5	Rung reached, error codes, worst crossing.	Written
Total	60		

Day 12 — Vertical sweeps

Today's single objective: Play the I/Ly starting row cleanly in C major, labels spoken, at rung 3.

The two rows where the shift arrives earlier and hurts more.

Row starting on I (Ionian) — C major, lowest position

Labels: **I > Ly > Lo > P**. Starting frets E-A-D-G: 8, 8, 9, 9. There is fret shift of +1 when crossing to the D string (Ly to Lo).

```
G|-----9-10-12-|
D|-----9-10-12-----|
A|-----8-10-12-----|
E|---8-10-12-----|
```

Notes: C D E | F G A | B C D | E F G

Row starting on Ly (Lydian) — C major, lowest position

Labels: **Ly > Lo > P > A**. Starting frets E-A-D-G: 1, 2, 2, 2. There is fret shift of +1 when crossing to the A string (Ly to Lo).

```
G|-----2--4--5-|
D|-----2--3--5-----|
A|-----2--3--5-----|
E|---1--3--5-----|
```

Notes: F G A | B C D | E F G | A B C

The same row an octave up — identical labels, identical shapes, new column:

```
G|-----14-16-17-|
D|-----14-15-17-----|
A|-----14-15-17-----|
E|---13-15-17-----|
```

The six-pass protocol. Apply it to every row, every time. Never skip passes one to three just because you feel ready.

1. Say the four labels out loud before touching the bass.
2. Say the four starting frets.
3. Air-finger the twelve notes without plucking.
4. Play one note per click, saying the note names.
5. Play again, saying only the chunk labels.
6. Play again in silence, hearing the labels internally.

Block	Min	Content	Standard to hit
Hands	3	Warm-up plus one pass of each shape at fret 7.	Relaxed
Naming	5	Maintenance: 12 random coordinates.	11 of 12
Shape	5	Passages E1-E3, rung 4, one pass each.	Clean
Main material	24	Six-pass protocol on today's row(s). Then tempo ladder rungs 1 to 3.	3 clean at rung 3
Exploration	13	E9 sequence in threes — applied to today's row.	Labels still spoken correctly
Repair	5	Isolate the worst crossing. Play only the last note of the old string and the first note of the new one, eight times.	No hesitation
Log	5	Rung reached, error codes, worst crossing.	Written
Total	60		

Day 13 — Vertical sweeps

Today's single objective: Play any of the seven starting rows on command, in C major, at rung 4.

All seven, called at random.

All seven rows in C major, at their lowest playable position. Do not memorise this table. Use it only to check yourself after you have generated a row from the labels.

Start	Labels	E	A	D	G	Shift at
M	M I Ly Lo	3-5-7	3-5-7	3-5-7	4-5-7	G string
I	I Ly Lo P	8-10-12	8-10-12	9-10-12	9-10-12	D string
Ly	Ly Lo P A	1-3-5	2-3-5	2-3-5	2-4-5	A string
Lo	Lo P A D	7-8-10	7-8-10	7-9-10	7-9-10	none
P	P A D M	0-1-3	0-2-3	0-2-3	0-2-4	none
A	A D M I	5-7-8	5-7-8	5-7-9	5-7-9	none
D	D M I Ly	10-12-13	10-12-14	10-12-14	10-12-14	none

The six-pass protocol. Apply it to every row, every time. Never skip passes one to three just because you feel ready.

1. Say the four labels out loud before touching the bass.
2. Say the four starting frets.
3. Air-finger the twelve notes without plucking.
4. Play one note per click, saying the note names.
5. Play again, saying only the chunk labels.
6. Play again in silence, hearing the labels internally.

Block	Min	Content	Standard to hit
Hands	3	Warm-up plus one pass of each shape at fret 7.	Relaxed
Naming	5	Maintenance: 12 random coordinates.	11 of 12
Shape	5	Passages E1-E3, rung 4, one pass each.	Clean
Main material	24	Random row called every 90 seconds. Six-pass protocol compressed to passes 1, 2 and 4. Ladder rungs 3 to 4.	3 clean at rung 4
Exploration	13	E10 sequence in fours — applied to today's row.	Labels still spoken correctly
Repair	5	Isolate the worst crossing. Play only the last note of the old string and the first note of the new one, eight times.	No hesitation
Log	5	Rung reached, error codes, worst crossing.	Written
Total	60		

Phase 3 exit standard: any of the seven rows, called at random, played correctly in C major at rung 4 with labels spoken, three times in a row without an error. You should now be generating rows from labels rather than recalling tab.

PHASE 4 — TURNAROUNDS AND THE FULL CYCLE

(Days 14-17)

Everything up to now moved in one direction. The turnaround is the moment where direction, label order and shift direction all reverse simultaneously. It deserves its own days.

The two turnarounds

G-string turn (top of the neck side). You have just finished three notes on the G string. Take **one more scale note** — the next one above — play it, and then come back down. Label order reverses. The shift now happens on the **Lo to Ly** crossing and moves your hand **down** one fret.

E-string turn (bottom side). You have just finished three descending notes on the E string. Take the **next scale note above** the note you stopped on, and ascend from there. This is why each complete cycle starts two scale notes higher than the last.

Day 14 — Turnarounds — checkpoint

Today's single objective: Execute the G-string turn without hesitation, in isolation.

The six-note bridge. This is the entire turnaround, stripped of everything around it.

```
G|---0--2--4--5--4--2-----|
D|-----5--3--2-|
A|-----|
E|-----|
```

Three ascending notes on G, the turn note, two descending notes on G, then the first descending group on D. Loop it. Nothing before it, nothing after it.

Block	Min	Content	Standard to hit
Hands	3	Warm-up.	Relaxed
Naming	5	Maintenance.	Clean
Checkpoint	12	Re-run the Phase 3 exit standard: three random rows at rung 4.	Passed
Bridge drill	20	Passage E15 looped. Rungs 1 to 4. Say “turn” aloud on the turn note for the first ten repetitions, then think it.	10 clean loops at rung 4
Reverse labels	8	Away from the bass: recite MILLPAD backwards from each of the seven labels.	No hesitation
Exploration	7	E16 two-string micro-Beast, up and back.	Clean
Log	5	Record.	Written
Total	60		

Day 15 — Turnarounds

Today's single objective: Execute both turnarounds and connect one full up-down-up cycle.

The E-string turn. Same logic, opposite end. The descent finishes low on the E string; the next scale note **above** that becomes the start of the new ascent.

```
G|-----|
D|-----|
A|-----3--5--7-|
E|---5--3--1--3--5--7-----|
```

Three descending notes on E, then the turn, then straight into the next ascent. The new ascent starts **two scale notes higher** than the previous one did.

Block	Min	Content	Standard to hit
Hands	3	Warm-up.	Relaxed
Naming	4	Maintenance.	Clean
G-turn	10	E15 at rungs 3 to 5.	3 clean at rung 5
E-turn	12	The E-string bridge above, looped. Rungs 1 to 4.	3 clean at rung 4
Join them	18	Sweep 1 up, G-turn, sweep 2 down, E-turn, sweep 3 up. Stop there. Rungs 1 to 3.	3 clean at rung 3
Exploration	8	E14 root-pedal Beast — reintroduces a tonal centre so the exercise stops sounding like an exercise.	Clean
Log	5	Record. Which turn is weaker?	Written
Total	60		

Day 16 — Full cycle

Today's single objective: Play the complete C-major traversal, open position to fret 12, without stopping.

Here is the whole thing. Six sweeps, three up and three down, covering the entire neck from open position to fret 12.

#	Dir	Labels (play order)	E	A	D	G	Notes (play order)
1	UP	P A D M	0-1-3	0-2-3	0-2-3	0-2-4	E F G A B C D E F G A B
2	DOWN	A P Lo Ly	5-3-1	5-3-2	5-3-2	5-4-2	C B A G F E D C B A G F
3	UP	M I Ly Lo	3-5-7	3-5-7	3-5-7	4-5-7	G A B C D E F G A B C D
4	DOWN	I M D A	8-7-5	8-7-5	9-7-5	9-7-5	E D C B A G F E D C B A
5	UP	Lo P A D	7-8-10	7-8-10	7-9-10	7-9-10	B C D E F G A B C D E F
6	DOWN	P Lo Ly I	12-10-8	12-10-8	12-10-9	12-10-9	G F E D C B A G F E D C

Sweep 1 — ASCENDING — P > A > D > M

```
G|-----0--2--4-|
D|-----0--2--3-----|
A|-----0--2--3-----|
E|---0--1--3-----|
```

Sweep 2 — DESCENDING — A > P > Lo > Ly

```
G|---5--4--2-----|
D|-----5--3--2-----|
A|-----5--3--2-----|
E|-----5--3--1-----|
```

Sweep 3 — ASCENDING — M > I > Ly > Lo

```
G|-----4--5--7-|
D|-----3--5--7-----|
A|-----3--5--7-----|
E|---3--5--7-----|
```

Sweep 4 — DESCENDING — I > M > D > A

```
G|---9--7--5-----|
D|-----9--7--5-----|
A|-----8--7--5-----|
E|-----8--7--5-----|
```

Sweep 5 — ASCENDING — Lo > P > A > D

```
G|-----7--9-10-|
D|-----7--9-10-----|
A|-----7--8-10-----|
E|---7--8-10-----|
```

Sweep 6 — DESCENDING — P > Lo > Ly > I

```
G|---12-10--9-----|
D|-----12-10--9-----|
A|-----12-10--8-----|
E|-----12-10--8-----|
```

The route card. Once you can play it, throw away the tab above and use only this. Cover all rows but the one you are on.

C MAJOR – FULL TRAVERSAL ROUTE CARD

1 UP E A D G
P A D M
013 023 023 024

2 DOWN G D A E
A P Lo Ly
542 532 532 531

3 UP E A D G
M I Ly Lo
357 357 357 457

4 DOWN G D A E
I M D A
975 975 875 875

5 UP E A D G
Lo P A D
7810 7810 7910 7910

6 DOWN G D A E
P Lo Ly I
12109 12109 12108 12108

Block	Min	Content	Standard to hit
Hands	3	Warm-up.	Relaxed
Naming	4	Maintenance.	Clean
Sweep isolation	10	Each of the six sweeps individually, twice, rung 4.	All six clean
Joining	22	Sweeps 1-2, then 1-2-3, then 1-2-3-4, adding one sweep only after three clean repetitions. Rung 1 to 2 for joined material.	At least sweeps 1-4 joined
Repair	8	Whichever join broke. Checkpoint restart only — never from the top.	Fixed
Exploration	8	E13 octave-displaced chunks.	Clean
Log	5	How far did you get before the first error?	Written
Total	60		

Day 17 — Full cycle

Today's single objective: The complete C-major traversal, both directions, at rung 5, from memory.

Today you add the **descent of the whole structure**: after sweep 6, turn around at the top and come all the way back down to open position, ending where you started. The labels run backwards for the entire return journey.

Total length: twelve sweeps, one hundred and forty-four notes plus turn notes. At rung 5 (40 bpm, two notes per click) that is roughly one minute forty of unbroken playing.

Block	Min	Content	Standard to hit
Hands	3	Warm-up.	Relaxed
Naming	4	Maintenance.	Clean
Up-neck run	15	Sweeps 1 to 6 unbroken, rungs 3 to 5.	3 clean at rung 5
Down-neck run	15	From the top back to open position, rungs 1 to 3.	3 clean at rung 3
Whole thing	12	Up and back, once, at rung 2. Do not chase tempo today.	One clean pass
Exploration	6	E12 diatonic triads — start hearing the harmony under the exercise.	Chord names spoken
Log	5	Record. This is the halfway mark; write down what has changed since Day 1.	Written
Total	60		

Phase 4 exit standard: the complete C-major traversal, open position to fret 12 and back, played once without stopping at rung 3 or better, with the state model (key, direction, string, chunk) recoverable if someone stops you at random.

PHASE 5 — TRANSPOSITION (Days 18-23)

Everything you have built is now key-agnostic in principle. This phase makes it key-agnostic in fact. One new key per day, in order of increasing notational awkwardness.

The seven-step key procedure. Do it on paper first, every time, before you play a note.

1. Write the seven notes of the key with correct spelling.
2. Number them one to seven.
3. Map the degrees to labels: 1=I, 2=D, 3=P, 4=Ly, 5=M, 6=A, 7=Lo.
4. Find the lowest in-key note on the E string.
5. Read its label, then the next three labels forwards — that is your row.
6. Write the four starting frets, adding one at the Ly-to-Lo crossing if it occurs.
7. Play it. Only then look at the printed tab to check.

If step 7 disagrees with your paper, **the paper is where the lesson is**. Find the disagreement before you play it again.

Day 18 — Transposition — G major

Today's single objective: Generate and play the full G-major traversal without reading tab first.

One sharp. The friendliest transposition and the one to build confidence on.

Notes: G A B C D E F#

Scale notes on the neck:

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	G	.	A	.	B	C	.	D	.	E	.	F#	G
D	D	.	E	.	F#	G	.	A	.	B	C	.	D
A	A	.	B	C	.	D	.	E	.	F#	G	.	A
E	E	.	F#	G	.	A	.	B	C	.	D	.	E

Label map:

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	I	.	D	.	P	Ly	.	M	.	A	.	Lo	I
D	M	.	A	.	Lo	I	.	D	.	P	Ly	.	M
A	D	.	P	Ly	.	M	.	A	.	Lo	I	.	D
E	A	.	Lo	I	.	D	.	P	Ly	.	M	.	A

Check your work against this only after you have generated it yourself:

#	Dir	Labels (play order)	E	A	D	G	Notes (play order)
1	UP	A D M I	0-2-3	0-2-3	0-2-4	0-2-4	E F# G A B C D E F# G A B
2	DOWN	D A P Lo	5-3-2	5-3-2	5-4-2	5-4-2	C B A G F# E D C B A G F#
3	UP	I Ly Lo P	3-5-7	3-5-7	4-5-7	4-5-7	G A B C D E F# G A B C D
4	DOWN	Ly I M D	8-7-5	9-7-5	9-7-5	9-7-5	E D C B A G F# E D C B A

Sweep 1 (ascending): A > D > M > I

```
G|-----0--2--4-|
D|-----0--2--4-|
A|-----0--2--3-|
E|---0--2--3-|
```

Sweep 2 (descending): D > A > P > Lo

```
G|---5--4--2-|
D|-----5--4--2-|
A|-----5--3--2-|
E|-----5--3--2-|
```

Sweep 3 (ascending): I > Ly > Lo > P

```
G|-----4--5--7-|
D|-----4--5--7-|
A|-----3--5--7-|
E|---3--5--7-|
```

Sweep 4 (descending): Ly > I > M > D

```
G|---9--7--5-|
D|-----9--7--5-|
A|-----9--7--5-|
E|-----8--7--5-|
```


Block	Min	Content	Standard to hit
Hands	3	Warm-up.	Relaxed
Naming	4	Maintenance, weighted toward the accidentals in today's key.	Clean
Paper	8	The seven-step key procedure, written out, before touching the bass.	Matches the printed table
Row generation	14	Generate and play all four no-shift rows in today's key from labels alone. Rungs 1 to 3.	3 clean at rung 3
Full traversal	16	Sweeps 1 to 4 joined, with both turnarounds. Rungs 1 to 3.	3 clean at rung 3
Interleave	10	Interleave: one sweep in today's key, one sweep in a previous key, alternating.	No key contamination
Log	5	Which accidental did your fingers default past? Note it.	Written
Total	60		

Day 19 — Transposition — F major

Today's single objective: Generate and play the full F-major traversal without reading tab first.

One flat. Your first B-flat, which sits where you expect B and breaks autopilot.

Notes: F G A Bb C D E

Scale notes on the neck:

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	G	.	A	Bb	.	C	.	D	.	E	F	.	G
D	D	.	E	F	.	G	.	A	Bb	.	C	.	D
A	A	Bb	.	C	.	D	.	E	F	.	G	.	A
E	E	F	.	G	.	A	Bb	.	C	.	D	.	E

Label map:

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	D	.	P	Ly	.	M	.	A	.	Lo	I	.	D
D	A	.	Lo	I	.	D	.	P	Ly	.	M	.	A
A	P	Ly	.	M	.	A	.	Lo	I	.	D	.	P
E	Lo	I	.	D	.	P	Ly	.	M	.	A	.	Lo

Check your work against this only after you have generated it yourself:

#	Dir	Labels (play order)	E	A	D	G	Notes (play order)
1	UP	Lo P A D	0-1-3	0-1-3	0-2-3	0-2-3	E F G A Bb C D E F G A Bb
2	DOWN	P Lo Ly I	5-3-1	5-3-1	5-3-2	5-3-2	C Bb A G F E D C Bb A G F
3	UP	D M I Ly	3-5-6	3-5-7	3-5-7	3-5-7	G A Bb C D E F G A Bb C D
4	DOWN	M D A P	8-6-5	8-7-5	8-7-5	9-7-5	E D C Bb A G F E D C Bb A

Sweep 1 (ascending): Lo > P > A > D

```
G|-----0--2--3-|
D|-----0--2--3-|
A|-----0--1--3-|
E|---0--1--3-|
```

Sweep 2 (descending): P > Lo > Ly > I

```
G|---5--3--2-----|
D|-----5--3--2-----|
A|-----5--3--1-----|
E|-----5--3--1-|
```

Sweep 3 (ascending): D > M > I > Ly

```
G|-----3--5--7-|
D|-----3--5--7-|
A|-----3--5--7-|
E|---3--5--6-----|
```

Sweep 4 (descending): M > D > A > P

```
G|---9--7--5-----|
D|-----8--7--5-----|
A|-----8--7--5-----|
E|-----8--6--5-|
```

Block	Min	Content	Standard to hit
Hands	3	Warm-up.	Relaxed
Naming	4	Maintenance, weighted toward the accidentals in today's key.	Clean
Paper	8	The seven-step key procedure, written out, before touching the bass.	Matches the printed table
Row generation	14	Generate and play all four no-shift rows in today's key from labels alone. Rungs 1 to 3.	3 clean at rung 3
Full traversal	16	Sweeps 1 to 4 joined, with both turnarounds. Rungs 1 to 3.	3 clean at rung 3
Interleave	10	Interleave: one sweep in today's key, one sweep in a previous key, alternating.	No key contamination
Log	5	Which accidental did your fingers default past? Note it.	Written
Total	60		

Day 20 — Transposition — D major

Today's single objective: Generate and play the full D-major traversal without reading tab first.

Two sharps. Two accidentals means two places your fingers will default wrongly.

Notes: D E F# G A B C#

Scale notes on the neck:

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	G	.	A	.	B	.	C#	D	.	E	.	F#	G
D	D	.	E	.	F#	G	.	A	.	B	.	C#	D
A	A	.	B	.	C#	D	.	E	.	F#	G	.	A
E	E	.	F#	G	.	A	.	B	.	C#	D	.	E

Label map:

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	Ly	.	M	.	A	.	Lo	I	.	D	.	P	Ly
D	I	.	D	.	P	Ly	.	M	.	A	.	Lo	I
A	M	.	A	.	Lo	I	.	D	.	P	Ly	.	M
E	D	.	P	Ly	.	M	.	A	.	Lo	I	.	D

Check your work against this only after you have generated it yourself:

#	Dir	Labels (play order)	E	A	D	G	Notes (play order)
1	UP	D M I Ly	0-2-3	0-2-4	0-2-4	0-2-4	E F# G A B C# D E F# G A B
2	DOWN	M D A P	5-3-2	5-4-2	5-4-2	6-4-2	C# B A G F# E D C# B A G F#
3	UP	Ly Lo P A	3-5-7	4-5-7	4-5-7	4-6-7	G A B C# D E F# G A B C# D
4	DOWN	Lo Ly I M	9-7-5	9-7-5	9-7-5	9-7-6	E D C# B A G F# E D C# B A

Sweep 1 (ascending): D > M > I > Ly

```
G|-----0--2--4-|
D|-----0--2--4-|
A|-----0--2--4-|
E|---0--2--3-----|
```

Sweep 2 (descending): M > D > A > P

```
G|---6--4--2-----|
D|-----5--4--2-----|
A|-----5--4--2-----|
E|-----5--3--2-|
```

Sweep 3 (ascending): Ly > Lo > P > A

```
G|-----4--6--7-|
D|-----4--5--7-|
A|-----4--5--7-|
E|---3--5--7-----|
```

Sweep 4 (descending): Lo > Ly > I > M

```
G|---9--7--6-----|
D|-----9--7--5-----|
A|-----9--7--5-----|
E|-----9--7--5-|
```

Block	Min	Content	Standard to hit
Hands	3	Warm-up.	Relaxed
Naming	4	Maintenance, weighted toward the accidentals in today's key.	Clean
Paper	8	The seven-step key procedure, written out, before touching the bass.	Matches the printed table
Row generation	14	Generate and play all four no-shift rows in today's key from labels alone. Rungs 1 to 3.	3 clean at rung 3
Full traversal	16	Sweeps 1 to 4 joined, with both turnarounds. Rungs 1 to 3.	3 clean at rung 3
Interleave	10	Interleave: one sweep in today's key, one sweep in a previous key, alternating.	No key contamination
Log	5	Which accidental did your fingers default past? Note it.	Written
Total	60		

Day 21 — Transposition — Bb major

Today's single objective: Generate and play the full Bb-major traversal without reading tab first.

Two flats — checkpoint day. Bass keys live here; this one earns its keep.

Notes: Bb C D Eb F G A

Scale notes on the neck:

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	G	.	A	Bb	.	C	.	D	Eb	.	F	.	G
D	D	Eb	.	F	.	G	.	A	Bb	.	C	.	D
A	A	Bb	.	C	.	D	Eb	.	F	.	G	.	A
E	.	F	.	G	.	A	Bb	.	C	.	D	Eb	.

Label map:

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	A	.	Lo	I	.	D	.	P	Ly	.	M	.	A
D	P	Ly	.	M	.	A	.	Lo	I	.	D	.	P
A	Lo	I	.	D	.	P	Ly	.	M	.	A	.	Lo
E	.	M	.	A	.	Lo	I	.	D	.	P	Ly	.

Check your work against this only after you have generated it yourself:

#	Dir	Labels (play order)	E	A	D	G	Notes (play order)
1	UP	M I Ly Lo	1-3-5	1-3-5	1-3-5	2-3-5	F G A Bb C D Eb F G A Bb C
2	DOWN	I M D A	6-5-3	6-5-3	7-5-3	7-5-3	D C Bb A G F Eb D C Bb A G
3	UP	Lo P A D	5-6-8	5-6-8	5-7-8	5-7-8	A Bb C D Eb F G A Bb C D Eb
4	DOWN	P Lo Ly I	10-8-6	10-8-6	10-8-7	10-8-7	F Eb D C Bb A G F Eb D C Bb

Sweep 1 (ascending): M > I > Ly > Lo

```
G|-----2--3--5-|
D|-----1--3--5-----|
A|-----1--3--5-----|
E|---1--3--5-----|
```

Sweep 2 (descending): I > M > D > A

```
G|---7--5--3-----|
D|-----7--5--3-----|
A|-----6--5--3-----|
E|-----6--5--3-|
```

Sweep 3 (ascending): Lo > P > A > D

```
G|-----5--7--8-|
D|-----5--7--8-----|
A|-----5--6--8-----|
E|---5--6--8-----|
```

Sweep 4 (descending): P > Lo > Ly > I

```
G|---10--8--7-----|
D|-----10--8--7-----|
A|-----10--8--6-----|
E|-----10--8--6-|
```

Block	Min	Content	Standard to hit
Hands	3	Warm-up.	Relaxed
Naming	4	Maintenance, weighted toward the accidentals in today's key.	Clean
Paper	8	The seven-step key procedure, written out, before touching the bass.	Matches the printed table
Row generation	14	Generate and play all four no-shift rows in today's key from labels alone. Rungs 1 to 3.	3 clean at rung 3
Full traversal	16	Sweeps 1 to 4 joined, with both turnarounds. Rungs 1 to 3.	3 clean at rung 3
Interleave	10	Re-run the Phase 4 exit standard in C major before starting the new key.	No key contamination
Log	5	Which accidental did your fingers default past? Note it.	Written
Total	60		

Day 22 — Transposition — A major

Today's single objective: Generate and play the full A-major traversal without reading tab first.

Three sharps. The point at which paper-first becomes non-negotiable.

Notes: A B C# D E F# G#

Scale notes on the neck:

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	.	G#	A	.	B	.	C#	D	.	E	.	F#	.
D	D	.	E	.	F#	.	G#	A	.	B	.	C#	D
A	A	.	B	.	C#	D	.	E	.	F#	.	G#	A
E	E	.	F#	.	G#	A	.	B	.	C#	D	.	E

Label map:

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	.	Lo	I	.	D	.	P	Ly	.	M	.	A	.
D	Ly	.	M	.	A	.	Lo	I	.	D	.	P	Ly
A	I	.	D	.	P	Ly	.	M	.	A	.	Lo	I
E	M	.	A	.	Lo	I	.	D	.	P	Ly	.	M

Check your work against this only after you have generated it yourself:

#	Dir	Labels (play order)	E	A	D	G	Notes (play order)
1	UP	M I Ly Lo	0-2-4	0-2-4	0-2-4	1-2-4	E F# G# A B C# D E F# G# A B
2	DOWN	I M D A	5-4-2	5-4-2	6-4-2	6-4-2	C# B A G# F# E D C# B A G# F#
3	UP	Lo P A D	4-5-7	4-5-7	4-6-7	4-6-7	G# A B C# D E F# G# A B C# D
4	DOWN	P Lo Ly I	9-7-5	9-7-5	9-7-6	9-7-6	E D C# B A G# F# E D C# B A

Sweep 1 (ascending): M > I > Ly > Lo

```
G|-----1--2--4-|
D|-----0--2--4-|
A|-----0--2--4-|
E|---0--2--4-|
```

Sweep 2 (descending): I > M > D > A

```
G|---6--4--2-|
D|-----6--4--2-|
A|-----5--4--2-|
E|-----5--4--2-|
```

Sweep 3 (ascending): Lo > P > A > D

```
G|-----4--6--7-|
D|-----4--6--7-|
A|-----4--5--7-|
E|---4--5--7-|
```

Sweep 4 (descending): P > Lo > Ly > I

```
G|---9--7--6-|
D|-----9--7--6-|
A|-----9--7--5-|
E|-----9--7--5-|
```


Block	Min	Content	Standard to hit
Hands	3	Warm-up.	Relaxed
Naming	4	Maintenance, weighted toward the accidentals in today's key.	Clean
Paper	8	The seven-step key procedure, written out, before touching the bass.	Matches the printed table
Row generation	14	Generate and play all four no-shift rows in today's key from labels alone. Rungs 1 to 3.	3 clean at rung 3
Full traversal	16	Sweeps 1 to 4 joined, with both turnarounds. Rungs 1 to 3.	3 clean at rung 3
Interleave	10	Interleave: one sweep in today's key, one sweep in a previous key, alternating.	No key contamination
Log	5	Which accidental did your fingers default past? Note it.	Written
Total	60		

Day 23 — Transposition — Eb major

Today's single objective: Generate and play the full Eb-major traversal without reading tab first.

Three flats. Awkward on purpose. If you can generate this one, you can generate all twelve.

Notes: Eb F G Ab Bb C D

Scale notes on the neck:

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	G	Ab	.	Bb	.	C	.	D	Eb	.	F	.	G
D	D	Eb	.	F	.	G	Ab	.	Bb	.	C	.	D
A	.	Bb	.	C	.	D	Eb	.	F	.	G	Ab	.
E	.	F	.	G	Ab	.	Bb	.	C	.	D	Eb	.

Label map:

Str	0	1	2	3	4	5	6	7	8	9	10	11	12
G	P	Ly	.	M	.	A	.	Lo	I	.	D	.	P
D	Lo	I	.	D	.	P	Ly	.	M	.	A	.	Lo
A	.	M	.	A	.	Lo	I	.	D	.	P	Ly	.
E	.	D	.	P	Ly	.	M	.	A	.	Lo	I	.

Check your work against this only after you have generated it yourself:

#	Dir	Labels (play order)	E	A	D	G	Notes (play order)
1	UP	D M I Ly	1-3-4	1-3-5	1-3-5	1-3-5	F G Ab Bb C D Eb F G Ab Bb C
2	DOWN	M D A P	6-4-3	6-5-3	6-5-3	7-5-3	D C Bb Ab G F Eb D C Bb Ab G
3	UP	Ly Lo P A	4-6-8	5-6-8	5-6-8	5-7-8	Ab Bb C D Eb F G Ab Bb C D Eb
4	DOWN	Lo Ly I M	10-8-6	10-8-6	10-8-6	10-8-7	F Eb D C Bb Ab G F Eb D C Bb

Sweep 1 (ascending): D > M > I > Ly

```
G|-----1--3--5-|
D|-----1--3--5-----|
A|-----1--3--5-----|
E|---1--3--4-----|
```

Sweep 2 (descending): M > D > A > P

```
G|---7--5--3-----|
D|-----6--5--3-----|
A|-----6--5--3-----|
E|-----6--4--3-|
```

Sweep 3 (ascending): Ly > Lo > P > A

```
G|-----5--7--8-|
D|-----5--6--8-----|
A|-----5--6--8-----|
E|---4--6--8-----|
```

Sweep 4 (descending): Lo > Ly > I > M

```
G|--10--8--7-----|
D|-----10--8--6-----|
A|-----10--8--6-----|
E|-----10--8--6-|
```

Block	Min	Content	Standard to hit
Hands	3	Warm-up.	Relaxed
Naming	4	Maintenance, weighted toward the accidentals in today's key.	Clean
Paper	8	The seven-step key procedure, written out, before touching the bass.	Matches the printed table
Row generation	14	Generate and play all four no-shift rows in today's key from labels alone. Rungs 1 to 3.	3 clean at rung 3
Full traversal	16	Sweeps 1 to 4 joined, with both turnarounds. Rungs 1 to 3.	3 clean at rung 3
Interleave	10	Interleave: one sweep in today's key, one sweep in a previous key, alternating.	No key contamination
Log	5	Which accidental did your fingers default past? Note it.	Written
Total	60		

Phase 5 exit standard: given any of the seven keys covered, generate the correct starting row on paper in under sixty seconds, then play the first four sweeps at rung 3 without reading tab.

PHASE 6 — FREE NAVIGATION (Days 24-26)

No new information from here on. What changes is that nothing is pre-selected: key, starting fret, direction and tempo all arrive at random, and you have to construct the answer live. This is the phase that converts a practised exercise into actual fretboard navigation.

Randomisation method. Roll or draw three things at the start of each attempt:

Draw	Range	Meaning
Key	any of the seven learned keys	parent scale
String and fret	any in-key location, frets 0-14	starting note
Direction	up or down	first sweep direction

If the drawn fret is not in the drawn key, redraw the fret. A chromatic note cannot begin a diatonic sweep — that is not a failure of the system, it is the definition of the system.

Day 24 — Free navigation

Today's single objective: Start a correct sweep from any in-key note on the E string, in any learned key, within ten seconds.

The start-anywhere worksheet. Fill it in mentally, in this order, every time. With practice the whole thing takes about four seconds.

Question	Example answer
What is the parent key?	D major
What note is at the chosen fret?	E string, fret 5 = A
Is it in the key?	Yes
Which degree?	Five
Which label?	Mixolydian
Which shape?	WS-WS
Which four-label row?	M - I - Ly - Lo
Where is the shift?	Last crossing, D to G string, +1 fret

That worked example, played:

```
G|-----6--7--9-|
D|-----5--7--9-----|
A|-----5--7--9-----|
E|---5--7--9-----|
```

Block	Min	Content	Standard to hit
Hands	3	Warm-up.	Relaxed
Naming	4	Maintenance across all four strings.	Clean
Worksheet drill	15	Random key and random E-string fret. Speak the whole worksheet aloud, then play the row. 20 attempts.	18 of 20 inside 10 sec
Silent worksheet	12	Same, but think the worksheet instead of speaking it. 20 attempts.	18 of 20 inside 6 sec
Full sweeps	16	From each drawn start, play two full sweeps with the turnaround. Rung 3.	3 clean
Exploration	6	E21 transposition passages, any two keys.	Generated, not read
Log	5	Average time to first note.	Written
Total	61		

Day 25 — Free navigation

Today's single objective: Start from any string, not just the E string, and run the cycle in both directions.

Starting on the A, D or G string means your first sweep is short — you have fewer strings above you before the turnaround. This is normal and it is the situation you will actually be in when using this material musically. **The label logic is unchanged.** You simply read fewer labels before turning.

Rule for a short sweep: read as many labels forwards as you have strings remaining, take the turn note, then read backwards.

Block	Min	Content	Standard to hit
Hands	3	Warm-up.	Relaxed
Naming	4	Maintenance.	Clean
Short sweeps	18	Random start on A, D or G string. Play up to the G string, turn, come back down to the E string, turn, continue. 15 attempts.	13 of 15 clean
Downward first	15	Random start, but descend first. The label track runs backwards from note one.	10 clean attempts
Long run	12	One continuous run: any key, any start, up to fret 15 and back down to open position.	One unbroken pass at rung 2
Exploration	8	E11 diatonic thirds in the drawn key.	Clean
Log	5	Record.	Written
Total	65		

Day 26 — Free navigation

Today's single objective: Full neck: open position to fret 17 and back, any key, no chart, no hesitation.

Today the upper neck comes in properly. Above fret 12 everything repeats an octave higher, so there is nothing new to learn — but the frets are narrower, the WS-WS stretch becomes easy, and your visual landmarks change. Use fret 12 as the mirror of the open string, fret 15 as the mirror of fret 3, fret 17 as the mirror of fret 5.

Block	Min	Content	Standard to hit
Hands	3	Warm-up in the upper register specifically.	Relaxed
Naming	6	Coordinates from fret 12 to fret 17 only.	18 of 20
Upper-neck rows	14	All seven rows starting above fret 12, in C then in one drawn key.	Clean at rung 4
Whole-neck run	20	Open position to fret 17 and back, one drawn key, unbroken.	Two clean passes at rung 3
Speed	7	Best key only. Push to rung 6 or 7 for three repetitions.	3 clean
Exploration	5	E10 sequence in fours, upper neck.	Clean
Log	5	Record top rung achieved.	Written
Total	60		

Phase 6 exit standard: drawn key, drawn starting note, played correctly within six seconds of the draw, run to fret 17 and back without stopping, and if halted at random you can state key, direction, string and chunk immediately.

PHASE 7 — MUSICAL INTEGRATION AND ASSESSMENT (Days 27-30)

The exercise has done its job when you stop needing it. These four days convert hand shapes into sound and then verify the whole month.

Day 27 — Integration

Today's single objective: Make each mode audible rather than theoretical.

Set up a drone: an open string, a held low note, a looper, a synth pad, anything. Then play only Beast chunks that fall under your hand, landing on the drone note every four bars.

The rule that makes this work: **emphasise the one note that distinguishes the mode from its nearest neighbour.** That note is doing all of the character work; everything else is shared.

Mode	Nearest neighbour	The note that does the work
Dorian	Aeolian	major 6th
Phrygian	Aeolian	flat 2nd
Lydian	Ionian	sharp 4th
Mixolydian	Ionian	flat 7th
Aeolian	Dorian	flat 6th
Locrian	Phrygian	flat 5th

Block	Min	Content	Standard to hit
Hands	3	Warm-up.	Relaxed
Maintenance	7	One full traversal in a drawn key at rung 4.	Clean
Drone work	25	Passages E17.2 through E17.7, one mode at a time, four minutes each with a drone.	You can hear the difference
A/B test	12	Alternate two bars of Dorian and two of Aeolian over the same drone. Then Phrygian against Aeolian.	You can name which is which blind
Exploration	8	E18-E20 riff seeds. Change the rhythm until one stops being an exercise.	One usable idea written down
Log	5	Which mode is easiest to hear? Which is hardest?	Written
Total	60		

Day 28 — Integration — checkpoint

Today's single objective: Use the material to write, not to practise.

Today you produce something. Take any drawn key, pick a mode, and build a bass line using only the root, third, fifth and characteristic note of that mode — plus Beast chunks as connective tissue between them.

This is the actual payoff of the whole month: the neck stops being a grid of positions and becomes a field you can move through in any direction while knowing where you are.

Block	Min	Content	Standard to hit
Hands	3	Warm-up.	Relaxed
Maintenance	8	Full traversal, drawn key, rung 5.	Clean
Checkpoint	14	Re-run the Phase 6 exit standard: three drawn starts, six-second limit.	Passed
Writing	25	Build three eight-bar bass lines in three different modes. Record them, however roughly.	Three lines recorded
Review	5	Listen back. Which one sounds most like a mode and least like a scale?	Identified
Log	5	Record.	Written
Total	60		

Day 29 — Consolidation

Today's single objective: Everything, at tempo, with no reference material within reach.

Put the manual in another room. Today is played entirely from the state model.

Block	Min	Content	Standard to hit
Hands	3	Warm-up.	Relaxed
All seven rows	12	C major, random order, rung 5.	All seven clean
All seven keys	20	One full up-and-back traversal in each key covered. Rung 4.	Seven traversals
Random draws	12	Ten drawn starts, six-second limit, two sweeps each.	9 of 10
Speed pass	8	Best key, push to your ceiling. Three clean repetitions at your top rung.	Ceiling identified
Log	5	Record the ceiling rung and the weakest key.	Written
Total	60		

Day 30 — Final assessment

Today's single objective: Prove the month.

Record the whole assessment on your phone. You will want the evidence, and the recording will show you things playing will not.

The eight tests. Score each pass or fail, honestly.

#	Test	Standard
1	Fretboard recall	30 random coordinates named, 28 correct, each under two seconds
2	Shape recall	All three shapes played from a called fret and string, 20 of 20
3	MILLPAD recall	Forwards and backwards from any of the seven starting labels, no hesitation
4	Row generation	Any label, any key: four labels, four frets and the shift location stated in under ten seconds
5	Full traversal	C major, open position to fret 12 and back, unbroken at rung 6
6	Transposition	Any of the seven keys, first four sweeps, unbroken at rung 4, no chart
7	Free start	Drawn key and drawn note, playing correctly within six seconds, ten times, nine correct
8	State recovery	Halted at random, state key, direction, string and chunk correctly, five of five

Block	Min	Content	Standard to hit
Hands	5	Full warm-up. Treat this as a performance.	Relaxed
Tests 1-3	12	Recall tests. Record.	Pass
Tests 4-6	20	Generation and traversal. Record.	Pass
Tests 7-8	13	Free start and state recovery. Have someone else halt you if possible.	Pass
Free play	5	No tests. Play something you like using what you now know.	Enjoy it
Log	5	Score all eight. Plan the next month from the failures.	Written
Total	60		

What to do with the result

Eight of eight. The exercise has done its job. Move to using it: harmonic minor and melodic minor use the identical machinery with a different parent collection, and the same seven-label cycle still applies — only the shape assignments change, because those scales have their semitones in different places. Working out that new shape table yourself is the best possible next project.

Six or seven. Repeat the phase that contains the failed test, five days, then re-test. Do not repeat the whole month.

Fewer than six. Almost always the cause is Phase 1. Go back to Days 1-5 and be honest about the two-second standard. Nothing downstream works without it.

The maintenance routine, from Day 31 onward

Ten minutes, three times a week, forever:

2 min	twenty random coordinates
2 min	three shapes, one pass each, at your top rung
4 min	one full traversal in a drawn key
2 min	one drawn start, worksheet silent, two sweeps

That is enough to hold everything you built. The thirty days are the acquisition; this is the retention.

PART FOUR — EXPLORATION PASSAGES

Scales practised as scales stay scales. These passages exist to force the same material through different mental frames, so the knowledge stops being a route and starts being a map. Each one is short on purpose. **The instruction under each passage matters more than the notes.**

Use them as prescribed in Part Three, or pull any of them out on a day when the main material feels stale. Never play one at a tempo where you cannot say the labels out loud while playing.

4.1 Shape-isolation passages (Days 1-7)

These contain no navigation at all. They exist to make the three hand shapes automatic so that later you can spend all your attention on *where you are* rather than *what your fingers are doing*.

E1 — WS-WS ladder, one string

Purpose: Wire the wide shape and, more importantly, wire the hand movement it requires.

```
G|-----|
D|-----|
A|-----|
E|--3--5--7--4--6--8--5--7--9--6--8-10--7--9-11--8-10-12-|
```

How to work it: Fret 3-5-7, then 4-6-8, then 5-7-9, and so on. Below fret 7 do **not** stretch: play finger 1, then finger 2 or 3, then finger 4, and let the whole hand travel with the line. Above fret 7 you may keep 1-3-4 planted. Stop instantly on any sharp pain, tingling or numbness.

Hammer it: Four ascents and four descents at 40 bpm, one note per click. Watch your thumb — if it is anchored and your wrist is bending to compensate, you are doing the wrong exercise.

E2 — 1-2-4 ladder, one string

Purpose: Wire the half-step-first shape. This is the one that feels cramped and is therefore skipped.

```
G|-----|
D|-----|
A|-----|
E|--5--6--8--6--7--9--7--8-10--8--9-11--9-10-12-10-11-13-|
```

How to work it: Fingers 1, 2, 4 on frets x , $x+1$, $x+3$. Say “half, whole” out loud on each repetition. The finger-2-to-finger-4 gap is the whole step; do not let finger 3 creep in.

Hammer it: Play it, then immediately play E3 without stopping. Your hand should feel the difference before your brain names it.

E3 — 1-3-4 ladder, one string

Purpose: Wire the half-step-last shape, and separate it from 1-2-4 by feel alone.

```
G|-----|
D|-----|
A|-----|
E|--5--7--8--6--8--9--7--9-10--8-10-11--9-11-12-10-12-13-|
```

How to work it: Fingers 1, 3, 4 on frets x , $x+2$, $x+3$. Say “whole, half”.

Hammer it: Alternate E2 and E3 with eyes closed. Have someone call a shape at random; you have two seconds to be playing it. This single drill removes about half of all Beast errors.

E4 — one shape across all four strings

Purpose: Prove that a shape is fret-column-shaped, not string-shaped.

```
G|-----5--7--8-----|
D|-----5--7--8-----8--7--5-----|
A|-----5--7--8-----8--7--5-----|
E|--5--7--8-----8--7--5-----|
```

How to work it: Same three frets, E to A to D to G and back down. Nothing changes but the string. Mute behind you: the string you just left must go silent.

Hammer it: Repeat with WS-WS and 1-2-4 at the same fret. Then repeat all three at fret 9, then at fret 2. Same shape, three neighbourhoods.

4.2 Beast-deforming passages (Days 8-17)

Now the shapes go back into the sweep, but the sweep gets bent. Every deformation below keeps the *labels* identical and changes the *order*, which is exactly what forces you to track state instead of replaying a memorised tape.

All of these are written in **C major from the open position** so you can compare them directly against the plain sweep. Transpose them freely once Day 18 arrives.

E5 — the plain sweep (reference)

Purpose: The control sample. Everything below is a deformation of this.

```
G|-----0--2--4-|
D|-----0--2--3-----|
A|-----0--2--3-----|
E|---0--1--3-----|
```

How to work it: Labels: P - A - D - M. Say them aloud, one per string.

Hammer it: Play it once before and once after every deformed version, so you always know what “undeformed” feels like.

E6 — pivot chunks (1-2-1-3)

Purpose: Breaks the linear-tape habit. You cannot play this on autopilot.

```
G|-----0--2--0--4-|
D|-----0--2--0--3-----|
A|-----0--2--0--3-----|
E|---0--1--0--3-----|
```

How to work it: Inside each three-note group, play note 1, note 2, back to note 1, then note 3. Sixteen notes across the four strings instead of twelve.

Hammer it: Say the chunk label on the *first* note of each group only. If you lose the label, your fingers are leading your brain — slow down until the brain is back in front.

E7 — broken chunks (1-3-2)

Purpose: Same shape, scrambled order. Attacks the assumption that the shape *is* the order.

```
G|-----0--4--2-|
D|-----0--3--2-----|
A|-----0--3--2-----|
E|---0--3--1-----|
```

How to work it: Play the lowest note, then the highest, then the middle. Keep the hand shape completely still — only the plucking order changes.

Hammer it: Do it descending as well: 3-1-2. Then alternate 1-3-2 and 3-1-2 between strings.

E8 — retrograde chunk order

Purpose: Chunks played G-D-A-E, but each chunk still ascends. Pure label tracking.

```
G|---0--2--4-----|
D|-----0--2--3-----|
A|-----0--2--3-----|
E|-----0--1--3-|
```

How to work it: You are reading MILLPAD **backwards** at the chunk level while each chunk still ascends. Labels come out as M - D - A - P.

Hammer it: Call the label *before* you play each group. If you can do this one clean, you have genuinely separated the label from the finger memory.

E9 — sequence in threes

Purpose: Melodic fluency. Converts the exercise into something usable in an actual line.

```

G|-----4-|
D|-----3---3-5-3-5-7-5-7---|
A|-----3---3-5-3-5-7-5-7---7-|
E|---0-1-3-1-3-5-3-5-7-5-7---7-|

```

How to work it: Groups 1-2-3, then 2-3-4, then 3-4-5, and so on up the scale. String assignment here is chosen for position rather than by the three-per-string rule — this is deliberate, so you learn that the scale exists independently of The Beast.

Hammer it: Play it once with Beast fingering (three per string) and once with position fingering. Notice they contain identical notes. That realisation is the point of the passage.

E10 — sequence in fours

Purpose: Four-note groups against a three-note-per-string layout creates a rolling cross-rhythm.

```

G|-----|
D|-----3---3-5---3-5-7-|
A|-----3---3-5---3-5-7-3-5-7-5-7-7-|
E|---0-1-3-5-1-3-5-7-3-5-7-5-7-7-|

```

How to work it: Groups of four ascending, shifting up one scale note each time. Count in 4 while the string crossings arrive in 3.

Hammer it: Accent the first of every four. Then accent the first of every three. Same notes, two completely different pieces of music.

E11 — diatonic thirds

Purpose: Thirds are where scales become harmony. This is the bridge to writing lines.

```

G|-----4-|
D|-----3---5-3-7-5-|
A|-----3---5-3-7-5-7-|
E|---0-3-1-5-3-7-5-7-|

```

How to work it: Note, skip one, note. Then move up one scale step and repeat.

Hammer it: Name the interval quality out loud as you play each pair: major third, minor third. The pattern of majors and minors *is* the key.

E12 — diatonic triads through the neck

Purpose: Turns the modal labels into chords, which is what they actually are.

```

G|-----4-|
D|-----3---5---3-7-5-|
A|-----3---5---3-7-5-7-|
E|---0-3-7-1-5-3-7-5-7-|

```

How to work it: Root, third, fifth from each successive scale degree. The chunk label and the triad quality line up: Ionian/Lydian/Mixolydian are major triads, Dorian/Phrygian/Aeolian are minor, Locrian is diminished.

Hammer it: Say the chord name, not the mode name: C, Dm, Em, F, G, Am, Bdim. You now have the entire harmony of the key under one exercise.

E13 — octave-displaced chunks

Purpose: Proves the “two strings up, two frets across” octave shape and destroys position tunnel vision.

```

G|-----2-4-5-|
D|-----2-3-5-|
A|-----0-2-3-|
E|---0-1-3-|

```

How to work it: Play a chunk on E, then the identical shape two strings up and two frets across on D. Then the A-string chunk and its copy on G.

Hammer it: Do it in reverse (high copy first). Then do it while naming the notes — you should hear the same three letters twice.

E14 — root-pedal Beast

Purpose: Forces a tonal centre onto the exercise, which is the beginning of modal *hearing*.

```

G|-----0-----2-----4-|
D|-----0-----2-----3-----|
A|-----0-----2-----3-----|
E|--8--0--8--1--8--3--8-----8-----8-----8-----8-----8-----|

```

How to work it: Alternate the low root with every note of the sweep. The pedal supplies the tonic that the raw exercise lacks.

Hammer it: Now move the pedal to a different scale degree — put it on A and play the same sweep. The identical notes will sound minor. That is relative modality demonstrated in ten seconds.

4.3 Turnaround passages (Days 12-17)

The turnaround is where nearly everyone breaks, because it is the one moment when both direction and label order flip at once. Practise the moment, not the journey to it.

E15 — the six-note G-string bridge

Purpose: Isolates the direction change to the smallest unit that still contains it.

```

G|---0--2--4--5--4--2-----|
D|-----5--3--2-|
A|-----|
E|-----|

```

How to work it: Three notes up on G, the turn note, two notes back down on G, then the first descending group on D. Nothing before it, nothing after it.

Hammer it: Ten repetitions as a loop at 40 bpm without stopping. Then say “turn” out loud exactly on the turn note. Then think it instead of saying it.

E16 — two-string micro-Beast

Purpose: A complete up-and-back with only one string crossing to manage.

```

G|-----|
D|-----|
A|-----0--2--3--3--2--0-----|
E|--0--1--3-----3--1--0-|

```

How to work it: E and A strings only, up and straight back down. Six notes each way.

Hammer it: Add the D string. Then the G string. Add one string per session and you will never experience the full sweep as difficult.

4.4 Modal-conversion passages (Days 24-30)

Everything so far treated the mode names as filenames for hand shapes. These passages convert them into music, which is the only reason to have learned any of it.

The recipe is always the same: **establish a tonal centre, then emphasise the one note that makes that mode different from its nearest neighbour.**

Mode	Compare against	The note that does the work	Try it over
Ionian	—	major 7th	C major
Dorian	Aeolian	major 6th	D minor / Dm7
Phrygian	Aeolian	flat 2nd	E / E minor drone
Lydian	Ionian	sharp 4th	F major
Mixolydian	Ionian	flat 7th	G7
Aeolian	Dorian	flat 6th	A minor
Locrian	Phrygian	flat 5th	Bm7b5

E17.2 — D Dorian statement (C major collection)

Purpose: Make D sound like home while playing nothing but C-major notes.


```

G|-----|
D|-----|
A|----12----14----14----|
E|--10----10----13----10-|

```

How to work it: Root, 5th, root, **characteristic tone**, 3rd, characteristic tone, root. The fourth and sixth notes are the ones that define the mode — lean on them and let the final D ring.

Hammer it: Loop a drone on D (an open string, a held note, a looper, a pedal on another instrument) and improvise using only the Beast chunks that fall under your hand. Land on D every four bars. Twelve repetitions. This is the exercise that makes the modes real.

E17.3 — E Phrygian statement (C major collection)

Purpose: Make E sound like home while playing nothing but C-major notes.

```

G|-----|
D|-----|
A|-----2-----|
E|--0-----0--1--3--1--0-|

```

How to work it: Root, 5th, root, **characteristic tone**, 3rd, characteristic tone, root. The fourth and sixth notes are the ones that define the mode — lean on them and let the final E ring.

Hammer it: Loop a drone on E (an open string, a held note, a looper, a pedal on another instrument) and improvise using only the Beast chunks that fall under your hand. Land on E every four bars. Twelve repetitions. This is the exercise that makes the modes real.

E17.4 — F Lydian statement (C major collection)

Purpose: Make F sound like home while playing nothing but C-major notes.

```

G|-----|
D|-----|
A|-----3-----2-----2-----|
E|--1-----1-----5-----1-|

```

How to work it: Root, 5th, root, **characteristic tone**, 3rd, characteristic tone, root. The fourth and sixth notes are the ones that define the mode — lean on them and let the final F ring.

Hammer it: Loop a drone on F (an open string, a held note, a looper, a pedal on another instrument) and improvise using only the Beast chunks that fall under your hand. Land on F every four bars. Twelve repetitions. This is the exercise that makes the modes real.

E17.5 — G Mixolydian statement (C major collection)

Purpose: Make G sound like home while playing nothing but C-major notes.

```

G|-----|
D|-----3-----3-----|
A|-----5-----|
E|--3-----3-----7-----3-|

```

How to work it: Root, 5th, root, **characteristic tone**, 3rd, characteristic tone, root. The fourth and sixth notes are the ones that define the mode — lean on them and let the final G ring.

Hammer it: Loop a drone on G (an open string, a held note, a looper, a pedal on another instrument) and improvise using only the Beast chunks that fall under your hand. Land on G every four bars. Twelve repetitions. This is the exercise that makes the modes real.

E17.6 — A Aeolian statement (C major collection)

Purpose: Make A sound like home while playing nothing but C-major notes.

```

G|-----|
D|-----|
A|-----7-----8-----8-----|
E|--5-----5-----8-----5-|

```

How to work it: Root, 5th, root, **characteristic tone**, 3rd, characteristic tone, root. The fourth and sixth notes are the ones that define the mode — lean on them and let the final A ring.

Hammer it: Loop a drone on A (an open string, a held note, a looper, a pedal on another instrument) and improvise

using only the Beast chunks that fall under your hand. Land on A every four bars. Twelve repetitions. This is the exercise that makes the modes real.

E17.7 — B Locrian statement (C major collection)

Purpose: Make B sound like home while playing nothing but C-major notes.

```
G|-----|
D|-----|
A|-----8-----8-----8-----|
E|---7-----7-----10-----7-|
```

How to work it: Root, 5th, root, **characteristic tone**, 3rd, characteristic tone, root. The fourth and sixth notes are the ones that define the mode — lean on them and let the final B ring.

Hammer it: Loop a drone on B (an open string, a held note, a looper, a pedal on another instrument) and improvise using only the Beast chunks that fall under your hand. Land on B every four bars. Twelve repetitions. This is the exercise that makes the modes real.

4.5 Riff seeds (any day you are bored)

You play bass in loud bands. These are not etudes; they are starting points. Each one is a diatonic cell built from Beast material with the low root doubled, sitting where a riff actually sits on the neck. Take one, change the rhythm, and it stops being an exercise.

E18 — Aeolian cell (A minor inside C major)

Purpose: The default rock/metal colour. Flat 6 is the character note.

```
G|-----|
D|-----|
A|-----7--5-----|
E|---5--5--7--5--5--8--5--7--5--5-----8--7-|
```

How to work it: Root heavy on the beat, upper notes as pickups. Palm-mute the root, let the top notes ring.

Hammer it: Move the identical shape to the Phrygian degree (E) without changing key. Same fingers, darker sound — that is the flat 2nd doing the work.

E19 — Phrygian cell (E Phrygian inside C major)

Purpose: The flat 2nd is the single most useful note in heavy music. Learn where it lives.

```
G|-----|
D|-----|
A|-----2--0-----|
E|---0--0--1--0--0--3--0--1--0--0-----3--1-|
```

How to work it: The half step between the root and the next note is the entire hook. Hit it hard, then fall back to the root.

Hammer it: Play the same cell in B-flat major starting on D. Then in G major starting on B. Same shape, three keys, ten seconds each.

E20 — Dorian cell (D Dorian inside C major)

Purpose: Dorian is Aeolian with a lifted 6th; it is the groove mode.

```
G|-----|
D|-----|
A|-----12-10-----|
E|---10-10-12-10-10-13-10-12-10-10-----13-12-|
```

How to work it: Let the major 6th ring against the root. Compare directly with E18 — the *only* difference between them is that one note.

Hammer it: Alternate one bar of E18 and one bar of E20. Hearing the 6th flip is the skill; the tab is just scaffolding.

4.6 Transposition passages (Days 18-23)

The same deformation drills, moved. Do not re-learn them — regenerate them. The point of Part One is that you should

be able to write these yourself from the state model alone.

E21 — G major from the fifth degree

Purpose: Same system, new letters. Generate it before you read it.

```
G|-----11-12-14-|
D|-----10-12-14-----|
A|-----10-12-14-----|
E|--10-12-14-----|
```

How to work it: Labels M - I - Ly - Lo. Starting frets by string, E to G: 10 10 10 11. Notes: G A B C D E F#.

Hammer it: Cover the tab. Say the four labels, say the four starting frets, then play. Only uncover the tab to check. If you can generate three keys this way, you can generate all twelve.

E21 — F major from the tonic

Purpose: Same system, new letters. Generate it before you read it.

```
G|-----2--3--5-|
D|-----2--3--5-----|
A|-----1--3--5-----|
E|--1--3--5-----|
```

How to work it: Labels I - Ly - Lo - P. Starting frets by string, E to G: 1 1 2 2. Notes: F G A Bb C D E.

Hammer it: Cover the tab. Say the four labels, say the four starting frets, then play. Only uncover the tab to check. If you can generate three keys this way, you can generate all twelve.

E21 — D major from the second degree

Purpose: Same system, new letters. Generate it before you read it.

```
G|-----12-14-16-|
D|-----12-14-16-----|
A|-----12-14-16-----|
E|--12-14-15-----|
```

How to work it: Labels D - M - I - Ly. Starting frets by string, E to G: 12 12 12 12. Notes: D E F# G A B C#.

Hammer it: Cover the tab. Say the four labels, say the four starting frets, then play. Only uncover the tab to check. If you can generate three keys this way, you can generate all twelve.

E21 — B-flat major from the fourth degree

Purpose: Same system, new letters. Generate it before you read it.

```
G|-----12-14-15-|
D|-----12-13-15-----|
A|-----12-13-15-----|
E|--11-13-15-----|
```

How to work it: Labels Ly - Lo - P - A. Starting frets by string, E to G: 11 12 12 12. Notes: Bb C D Eb F G A.

Hammer it: Cover the tab. Say the four labels, say the four starting frets, then play. Only uncover the tab to check. If you can generate three keys this way, you can generate all twelve.

PART FIVE — TROUBLESHOOTING AND LOGS

5.1 Error codes

Diagnose before you repeat. Undiagnosed repetition installs the error permanently.

Code	Error type	Typical example
N	Note error	Played F-sharp where the key wants F
C	Chunk error	Used 1-3-4 where the label called for 1-2-4
S	String error	Crossed to the G string after two notes instead of three
D	Direction error	Kept ascending after the turn note
T	Turnaround error	Repeated the last note instead of taking the next scale note
F	Fret-column error	Forgot the +1 at the Ly-to-Lo crossing
R	Rhythm error	Hesitation at a string crossing
M	Muting error	The string you left is still ringing
X	Tension error	Wrist locked, thumb clamped, or a painful stretch

The repair protocol

1. **Stop immediately.** Do not play through it.
2. **Name the code.** Out loud.
3. **State the intended note and label.**
4. Play the failed chunk alone, correctly, **five times**.
5. Add the chunk **before** it. Play both, **three times**.
6. Add the chunk **after** it. Play all three, **twice**.
7. Resume at least **one tempo rung slower**.
8. If the same code appears three sessions running, that is a Phase problem, not a repetition problem. Go back to the phase that teaches it.

Checkpoint restarts

Never restart from the beginning. Restart one checkpoint before the error.

Checkpoint A	E-string start of a sweep
Checkpoint B	entry to the D string
Checkpoint C	G-string turnaround
Checkpoint D	entry to the A string on the way down
Checkpoint E	E-string turnaround

5.2 The eleven common confusions

1. **“Am I playing E Phrygian when I play E-F-G?”** No. You are playing a Phrygian-shaped chunk. A mode needs a perceived tonic, which needs harmonic or rhythmic emphasis. Three passing notes cannot create one. Part Four section 4.4 is where you convert one into the other.
2. **“If I start on A, has the key changed to A minor?”** The note collection has not changed at all — C major and A Aeolian contain the same seven notes. Whether a listener hears C or A as home depends on emphasis and harmony, not on which note your first finger happened to land on.
3. **“1-2-4 and 1-3-4 look swapped in what I read.”** They probably were. Fingers 1, 2 and 4 in a one-finger-per-fret hand sit on frets x, x+1, x+3. Fingers 1, 3 and 4 sit on x, x+2, x+3. Derive it from the fingers and you cannot get it wrong.
4. **“Why is a descending group that sounds C-B-A labelled Aeolian?”** Because the label describes the shape written low to high. A-B-C is W-H, which is 1-3-4, which is Aeolian. You are playing an Aeolian shape backwards.
5. **“I keep forgetting the Ly-to-Lo shift.”** Say the memory phrase before every sweep: *after three wides, the index steps forward*. Then check: if your starting label is Lo, P, A or D, there is no shift in your four strings at all, so stop worrying about it on those rows.
6. **“The WS-WS stretch hurts.”** Then you are stretching where you should be shifting. Below about fret 7, play finger

1, then finger 2 or 3, then finger 4, and move the whole hand with the line. Or take the same material higher up the neck where the frets are closer. Pain is a technique signal, not a toughness test.

7. “I lose my place halfway through.” You are tracking fret numbers instead of state. Reduce to the four variables — key, direction, string, chunk — and say them at every string crossing. Twelve numbers is unmemorable; four variables is trivial.

8. “The tab is overwhelming.” Stop reading tab. Use a route card (5.4), cover every line but the one you are on, and generate the rest. The tab in this manual exists to check your generation, not to be read while playing.

9. “String crossings are where I rush.” Practise only the crossing: the last note of the old string and the first note of the new one, eight times at 40 bpm, then restore the full chunks.

10. “Open strings break the pattern.” They do not. Treat an open string as an imaginary first finger: 0-1-3 is a 1-2-4 chunk, 0-2-3 is a 1-3-4, 0-2-4 is a WS-WS. The fingering changes; the intervals do not.

11. “I can play it but I can’t transpose it.” Then you memorised tab, which is exactly what this manual is designed to prevent. Go back to Phase 5 and do the seven-step key procedure **on paper before playing**, every day, for a week. There is no shortcut past that step.

5.3 Hands and injury

Stop the session — not “ease off”, stop — on any of:

- Sharp pain anywhere in the hand, wrist or forearm
- Tingling or numbness in any finger
- A wrist that has to bend sharply to reach a note
- A thumb pressing hard enough to leave a mark on the neck

Structural rules that prevent nearly all of it:

- Fret immediately behind the fret wire, so you need less pressure.
- Release pressure between notes rather than gripping continuously.
- Keep the wrist close to neutral; move the arm to keep it there.
- A one-finger-per-fret hand covers **four** frets, not five. WS-WS spans five. Therefore WS-WS requires either a shift or a higher position. There is no third option.
- Mute what you are not playing. Muting is half of bass technique and it is invisible in tab.

5.4 Route cards

Play from these, not from tab. Cover every row but the one you are on.

C MAJOR – FULL TRAVERSAL				
1 UP	E	A	D	G
	P	A	D	M
	0-1-3	0-2-3	0-2-3	0-2-4
2 DOWN	G	D	A	E
	A	P	Lo	Ly
	5-4-2	5-3-2	5-3-2	5-3-1
3 UP	E	A	D	G
	M	I	Ly	Lo
	3-5-7	3-5-7	3-5-7	4-5-7
4 DOWN	G	D	A	E
	I	M	D	A
	9-7-5	9-7-5	8-7-5	8-7-5
5 UP	E	A	D	G
	Lo	P	A	D
	7-8-10	7-8-10	7-9-10	7-9-10
6 DOWN	G	D	A	E
	P	Lo	Ly	I
	12-10-9	12-10-9	12-10-8	12-10-8

Blank route card — fill one in for every key you work:

KEY: _____ NOTES: _____

#	DIR	STR	_____	_____	_____
1	UP	LAB	_____	_____	_____
			_____	_____	_____
2	DOWN	STR	_____	_____	_____
		LAB	_____	_____	_____
			_____	_____	_____
3	UP	STR	_____	_____	_____
		LAB	_____	_____	_____
			_____	_____	_____
4	DOWN	STR	_____	_____	_____
		LAB	_____	_____	_____
			_____	_____	_____

5.5 Logs

Daily log

Photocopy or redraw this. One line per day, thirty lines. The log is not administration; it is how you find out that your Thursday sessions are always worse than your Tuesday ones.

DAY ____ DATE _____ KEY _____ RANGE (frets) _____

Top tempo rung reached	_____
Clean repetitions at it	_____
Error codes seen	_____
Worst single moment	_____
Naming-drill score	_____ / _____
Time to first note (draws)	_____ sec
Body check (hands, wrist)	_____
One thing to fix tomorrow	_____

Weekly review

WEEK ____ DAYS ____ SESSIONS COMPLETED ____ / 7

Most repeated error code	_____
Which phase standard did I pass?	_____
Which key is weakest?	_____
Which string is slowest to name?	_____
Tempo rung, start of week	_____
Tempo rung, end of week	_____
Am I generating or recalling?	_____
Adjustment for next week	_____

Advancement rules

- **Advance the range** only when the current section plays three times with no note, direction or turnaround error.
- **Advance the tempo** only when the current rung is rhythmically even and the hands are relaxed. Correct but tense is not a pass.
- **Advance the phase** only on the printed phase exit standard. Repeating a phase is normal and costs you nothing; arriving at Phase 6 on a broken Phase 2 costs you the month.

5.6 The one-page cheat sheet

Cut this out. Tape it to your amp.

THE THREE SHAPES

WS-WS = x, x+2, x+4 = W-W = M I Ly
1-2-4 = x, x+1, x+3 = H-W = Lo P
1-3-4 = x, x+2, x+3 = W-H = A D

THE MASTER CYCLE

M - I - Ly - Lo - P - A - D - repeat
<- 3 wide -> <-2 H-W-> <-2 W-H->

DEGREES

1 I 2 D 3 P 4 Ly 5 M 6 A 7 Lo

ASCENDING

1. Find the starting note, its degree, its label.
2. Read four labels FORWARDS. One per string, E to G.
3. Three notes per string.
4. Shift +1 fret ONLY at Ly -> Lo.

G-STRING TURN

Take the next higher in-key note. Play it. Reverse.

DESCENDING

Read four labels BACKWARDS. Shift -1 fret at Lo -> Ly.

E-STRING TURN

Take the next higher in-key note. Reverse again.
Each full cycle starts TWO scale notes higher than the last.

TRACK ONLY FOUR THINGS

KEY - DIRECTION - STRING - CHUNK

WHY IT WORKS

3 scale notes = a 4th = one string. So the hand stays put.
Except F to B in C major, which is a tritone. That is the +1 fret.

ERROR REPAIR

Stop. Name the code. Chunk x5. Add previous x3. Add next x2. One rung slower.

IF IT HURTS

Shift the hand, or move up the neck. Never stretch through pain.

Generated for thirty days of one-hour sessions. Every chart, tab and table in this manual was computed from the rules in Part One rather than transcribed, so the twelve keys are internally consistent with each other and with the system. If you find a disagreement between a chart and the rule that should have produced it, trust the rule — and then work out which of you is wrong. That process is the skill this whole manual is trying to build.

APPENDIX A — EXTENDING TO FIVE STRINGS

A five-string bass tuned B-E-A-D-G changes nothing structural. The low B is a perfect fourth below the E, exactly like every other adjacent pair, so the entire argument in Part One survives untouched:

- Three scale notes per string is still a fourth, still cancelled by the tuning.
- The Ly-to-Lo tritone is still the only place the hand moves.
- MILLPAD is still the cycle, still 3-2-2.

The **only** difference is that a sweep now contains **five chunks and fifteen notes** instead of four and twelve, so you read five labels forwards instead of four. This means the Ly-to-Lo boundary now falls inside your sweep in **five** of the seven rows rather than three, so the shift comes up more often.

A.1 The seven rows on five strings

Start label	B	E	A	D	G	Fret columns	Shift at
M	M	I	Ly	Lo	P	x x x x+1 x+1	A to D
I	I	Ly	Lo	P	A	x x x+1 x+1 x+1	E to A
Ly	Ly	Lo	P	A	D	x x+1 x+1 x+1 x+1	B to E
Lo	Lo	P	A	D	M	x x x x x	none
P	P	A	D	M	I	x x x x x	none
A	A	D	M	I	Ly	x x x x x	none
D	D	M	I	Ly	Lo	x x x x x+1	D to G

Only the rows starting on **P, A and D** avoid the shift entirely.

A.2 Worked five-string sweeps in C major

Row starting on M (Mixolydian) — C major, fifteen notes

Labels: M > I > Ly > Lo > P

```
G|-----9--10--12-|
D|-----9--10--12-|
A|-----8--10--12-|
E|-----8--10--12-|
B|----8--10--12-|
```

Row starting on P (Phrygian) — C major, fifteen notes

Labels: P > A > D > M > I

```
G|-----5---7---9-|
D|-----5---7---9-|
A|-----5---7---8-|
E|-----5---7---8-|
B|----5---6---8-|
```

Row starting on I (Ionian) — C major, fifteen notes

Labels: I > Ly > Lo > P > A

```
G|-----2---4---5-|
D|-----2---3---5-|
A|-----2---3---5-|
E|-----1---3---5-|
B|----1---3---5-|
```

A.3 What actually changes in practice

Turnarounds get further apart. Fifteen notes between direction changes instead of twelve. The state model matters more, not less.

The B string is unfamiliar territory for naming. Add it to the Phase 1 naming drills: its natural notes are B=0, C=1, D=3, E=5, F=6, G=8, A=10, B=12. If you are working the thirty-day programme on a five-string, give Day 1 to the B

string and push the other strings back a day; run Phase 1 over six days rather than five.

The low register is muddier. Chunks below about the fifth fret of the B string will not speak clearly on most rigs. Practise them for the map, not for the tone, and do your musical work higher up.

Six strings. A low-B, high-C six-string is still all fourths, so a sweep is six chunks and eighteen notes and you read six labels. The system does not care how many strings you own; it only cares that they are tuned in fourths.

APPENDIX B — BEYOND THE MAJOR SCALE

Once the machinery is understood rather than memorised, it transfers to any seven-note scale. The rules that carry over unchanged are:

1. Three notes per string, four (or five) strings, then turn.
2. Each string crossing advances **three scale degrees**.
3. The fret shift on any crossing equals **(the size of that three-degree jump) minus five semitones**.

What changes is that non-major scales contain jumps other than fourths and tritones, so the shape table gets bigger and the shifts land in different places. The tables below are computed from rule 3. Work through one of them by hand before you trust it — that exercise is worth more than the table itself.

B.1 Major, for comparison

Degree	Semitones in the group	Fret offsets	Shape	Next chunk is +3 degrees	Fret shift
1	2, 2	0, +2, +4	WS-WS (1-3-4 / shift)	5 semitones	same column
2	2, 1	0, +2, +3	1-3-4	5 semitones	same column
3	1, 2	0, +1, +3	1-2-4	5 semitones	same column
4	2, 2	0, +2, +4	WS-WS (1-3-4 / shift)	6 semitones	+1 fret
5	2, 2	0, +2, +4	WS-WS (1-3-4 / shift)	5 semitones	same column
6	2, 1	0, +2, +3	1-3-4	5 semitones	same column
7	1, 2	0, +1, +3	1-2-4	5 semitones	same column

Six perfect fourths and one tritone. Three shapes. One shift. This is why the major scale is where you learn the system.

B.2 Harmonic minor

Intervals from the tonic: 0, 2, 3, 5, 7, 8, 11

Degree	Semitones in the group	Fret offsets	Shape	Next chunk is +3 degrees	Fret shift
1	2, 1	0, +2, +3	1-3-4	5 semitones	same column
2	1, 2	0, +1, +3	1-2-4	5 semitones	same column
3	2, 2	0, +2, +4	WS-WS (1-3-4 / shift)	5 semitones	same column
4	2, 1	0, +2, +3	1-3-4	6 semitones	+1 fret
5	1, 3	0, +1, +4	1-2-4+ (x, x+1, x+4) — wide	5 semitones	same column
6	3, 1	0, +3, +4	1-4-4+ (x, x+3, x+4) — wide	6 semitones	+1 fret
7	1, 2	0, +1, +3	1-2-4	4 semitones	-1 fret

B.3 Melodic minor (ascending)

Intervals from the tonic: 0, 2, 3, 5, 7, 9, 11

Degree	Semitones in the group	Fret offsets	Shape	Next chunk is +3 degrees	Fret shift
1	2, 1	0, +2, +3	1-3-4	5 semitones	same column
2	1, 2	0, +1, +3	1-2-4	5 semitones	same column
3	2, 2	0, +2, +4	WS-WS (1-3-4 / shift)	6 semitones	+1 fret
4	2, 2	0, +2, +4	WS-WS (1-3-4 / shift)	6 semitones	+1 fret
5	2, 2	0, +2, +4	WS-WS (1-3-4 / shift)	5 semitones	same column
6	2, 1	0, +2, +3	1-3-4	5 semitones	same column
7	1, 2	0, +1, +3	1-2-4	4 semitones	-1 fret

B.4 Harmonic major

Intervals from the tonic: 0, 2, 4, 5, 7, 8, 11

Degree	Semitones in the group	Fret offsets	Shape	Next chunk is +3 degrees	Fret shift
1	2, 2	0, +2, +4	WS-WS (1-3-4 / shift)	5 semitones	same column
2	2, 1	0, +2, +3	1-3-4	5 semitones	same column
3	1, 2	0, +1, +3	1-2-4	4 semitones	-1 fret
4	2, 1	0, +2, +3	1-3-4	6 semitones	+1 fret
5	1, 3	0, +1, +4	1-2-4+ (x, x+1, x+4) — wide	5 semitones	same column
6	3, 1	0, +3, +4	1-4-4+ (x, x+3, x+4) — wide	6 semitones	+1 fret
7	1, 2	0, +1, +3	1-2-4	5 semitones	same column

B.5 Reading those tables

Every figure below is computed from rule 3 above, not asserted. “Crossings that shift” counts, out of the seven crossings in a full cycle, how many require the hand to move a fret.

Scale	Distinct shapes needed	Crossings with no shift	Crossings that shift	Where they shift
Major (Ionian parent)	3	6	1	deg 4 (+1)
Harmonic minor	5	4	3	deg 4 (+1), deg 6 (+1), deg 7 (-1)
Melodic minor (ascending)	3	4	3	deg 3 (+1), deg 4 (+1), deg 7 (-1)
Harmonic major	5	4	3	deg 3 (-1), deg 4 (+1), deg 6 (+1)

The major scale is the outlier, and that is exactly why it is where you learn the system: **three shapes, one shift**. Nothing else in common use is that tidy.

Harmonic minor contains an augmented second between degrees 6 and 7, so a chunk spanning it needs a three-fret gap. More shapes, and three of the seven crossings move. Playable, genuinely harder, and not something to attempt before Day 30.

Melodic minor ascending is the most useful next step despite also having three shifting crossings, because its shapes stay narrow — no augmented second means no three-fret gaps, so your hand never has to do anything it has not already done in the major scale. If you take one non-major scale through this system next, take this one.

Harmonic major is included to show that the method generalises, not because you need it. Note that one of its shifts goes **downward**, which is a good test of whether you have understood rule 3 or merely memorised “plus one at Ly to Lo”.

A useful way to read all four tables at once: **the shift is never a special rule. It is always just the gap between what the music demands and the five semitones your tuning gives you**. In the major scale that gap is non-zero exactly once. In these scales it is non-zero more often. The principle has not changed at all.

B.6 Pentatonics

A five-note scale breaks the neat arithmetic, because three notes per string advances three degrees out of five rather than out of seven, so the cycle closes after five chunks instead of seven. Everything else still works. The minor pentatonic (0, 3, 5, 7, 10) gives:

Degree	Semitones in the group	Fret offsets	Jump of +3 degrees	Fret shift
1	3, 2	0, +3, +5	7 semitones	+2 fret
2	2, 2	0, +2, +4	7 semitones	+2 fret
3	2, 3	0, +2, +5	7 semitones	+2 fret
4	3, 2	0, +3, +5	8 semitones	+3 fret
5	2, 3	0, +2, +5	7 semitones	+2 fret

Three-notes-per-string pentatonic is a wide, awkward, and extremely useful thing to own. It is not part of the thirty days. It is the obvious project for the month after.

APPENDIX C — THIRTY DAYS AT A GLANCE

Print this page and put it where you practise. One line per day. Tick the box only when the standard is genuinely met.

Done	Day	Phase	Main work	Standard
[]	1	Foundation	E string names + WS-WS shape	20 coordinates, 18 correct, under 2 sec
[]	2	Foundation	A string names + 1-2-4 shape	20 coordinates, 18 correct
[]	3	Foundation	D string names + 1-3-4 shape	20 coordinates, 18 correct
[]	4	Foundation	G string names + WS-WS review	20 coordinates, 18 correct
[]	5	Foundation	Whole neck + shape switching	30 coordinates, 27 correct; 20 of 20 shape calls
[]	6	Linking	MILLPAD as a three-way lookup table	28 of 30 mixed prompts
[]	7	Linking	Labels attached to C-major degrees on the neck (checkpoint)	Phase 1 standard re-passed
[]	8	Linking	Any single chunk, any string, on command	27 of 30 in under 3 sec
[]	9	Sweeps	Rows starting Lo and P (no shift)	3 clean at rung 3
[]	10	Sweeps	Rows starting A and D (no shift)	3 clean at rung 3
[]	11	Sweeps	Row starting M (shift at the last crossing)	3 clean at rung 3
[]	12	Sweeps	Rows starting I and Ly (early shift)	3 clean at rung 3
[]	13	Sweeps	All seven rows at random	3 clean at rung 4
[]	14	Turnarounds	G-string turn in isolation (checkpoint)	10 clean loops at rung 4
[]	15	Turnarounds	Both turns, one full up-down-up	3 clean at rung 3
[]	16	Full cycle	C major, open position to fret 12	Sweeps 1-4 joined
[]	17	Full cycle	C major up and back, from memory	One clean pass at rung 2+
[]	18	Transposition	G major	3 clean at rung 3, generated not read
[]	19	Transposition	F major	3 clean at rung 3
[]	20	Transposition	D major	3 clean at rung 3
[]	21	Transposition	B-flat major (checkpoint)	Phase 4 standard re-passed
[]	22	Transposition	A major	3 clean at rung 3
[]	23	Transposition	E-flat major	Paper generation under 60 sec
[]	24	Free navigation	Any key, any E-string start	18 of 20 within 10 sec
[]	25	Free navigation	Any string, either direction first	13 of 15 clean
[]	26	Free navigation	Whole neck, open position to fret 17	2 clean passes at rung 3
[]	27	Integration	Modes as sound, over a drone	Can name modes blind
[]	28	Integration	Write three bass lines (checkpoint)	Three lines recorded
[]	29	Consolidation	All rows, all keys, no reference material	Seven traversals clean
[]	30	Assessment	The eight tests	Six of eight minimum

If you miss a day, do not double up the next day. Repeat the missed day and shift everything back. The programme is sequential by design; a compressed day teaches less than a delayed one.

If you fail a standard, repeat that day. Repeating is expected — most people repeat two or three days across the month, usually in Phase 1 and Phase 5. Thirty-three days to a real result beats thirty days to a shaky one.

APPENDIX D — ROUTE CARDS FOR THE SEVEN PRACTICE KEYS

Play from these, not from tablature. Each card is the whole traversal in three lines per sweep: which string, which label, which frets. Cover every sweep but the one you are on.

The purpose of a route card is that it gives you **less** information than tab. Tab tells your fingers what to do; a route card makes your brain do the work and only checks the answer. If you find you can play from tab but not from a route card, you have memorised rather than learned, and Phase 5 is where to go back to.

C MAJOR	C	D	E	F	G	A	B
1 UP	E		A		D		G
	P		A		D		M
	0-1-3		0-2-3		0-2-3		0-2-4
2 DOWN	G		D		A		E
	A		P		Lo		Ly
	5-4-2		5-3-2		5-3-2		5-3-1
3 UP	E		A		D		G
	M		I		Ly		Lo
	3-5-7		3-5-7		3-5-7		4-5-7
4 DOWN	G		D		A		E
	I		M		D		A
	9-7-5		9-7-5		8-7-5		8-7-5
5 UP	E		A		D		G
	Lo		P		A		D
	7-8-10		7-8-10		7-9-10		7-9-10
6 DOWN	G		D		A		E
	P		Lo		Ly		I
	12-10-9		12-10-9		12-10-8		12-10-8

G MAJOR	G	A	B	C	D	E	F#
1 UP	E		A		D		G
	A		D		M		I
	0-2-3		0-2-3		0-2-4		0-2-4
2 DOWN	G		D		A		E
	D		A		P		Lo
	5-4-2		5-4-2		5-3-2		5-3-2
3 UP	E		A		D		G
	I		Ly		Lo		P
	3-5-7		3-5-7		4-5-7		4-5-7
4 DOWN	G		D		A		E
	Ly		I		M		D
	9-7-5		9-7-5		9-7-5		8-7-5
5 UP	E		A		D		G
	P		A		D		M
	7-8-10		7-9-10		7-9-10		7-9-11
6 DOWN	G		D		A		E
	A		P		Lo		Ly
	12-11-9		12-10-9		12-10-9		12-10-8

F MAJOR F G A Bb C D E

1 UP	E	A	D	G
	Lo	P	A	D
	0-1-3	0-1-3	0-2-3	0-2-3
2 DOWN	G	D	A	E
	P	Lo	Ly	I
	5-3-2	5-3-2	5-3-1	5-3-1
3 UP	E	A	D	G
	D	M	I	Ly
	3-5-6	3-5-7	3-5-7	3-5-7
4 DOWN	G	D	A	E
	M	D	A	P
	9-7-5	8-7-5	8-7-5	8-6-5
5 UP	E	A	D	G
	Ly	Lo	P	A
	6-8-10	7-8-10	7-8-10	7-9-10
6 DOWN	G	D	A	E
	Lo	Ly	I	M
	12-10-9	12-10-8	12-10-8	12-10-8

D MAJOR D E F# G A B C#

1 UP	E	A	D	G
	D	M	I	Ly
	0-2-3	0-2-4	0-2-4	0-2-4
2 DOWN	G	D	A	E
	M	D	A	P
	6-4-2	5-4-2	5-4-2	5-3-2
3 UP	E	A	D	G
	Ly	Lo	P	A
	3-5-7	4-5-7	4-5-7	4-6-7
4 DOWN	G	D	A	E
	Lo	Ly	I	M
	9-7-6	9-7-5	9-7-5	9-7-5
5 UP	E	A	D	G
	A	D	M	I
	7-9-10	7-9-10	7-9-11	7-9-11
6 DOWN	G	D	A	E
	D	A	P	Lo
	12-11-9	12-11-9	12-10-9	12-10-9

BB MAJOR Bb C D Eb F G A

1 UP	E	A	D	G
	M	I	Ly	Lo
	1-3-5	1-3-5	1-3-5	2-3-5
2 DOWN	G	D	A	E
	I	M	D	A
	7-5-3	7-5-3	6-5-3	6-5-3
3 UP	E	A	D	G
	Lo	P	A	D
	5-6-8	5-6-8	5-7-8	5-7-8
4 DOWN	G	D	A	E
	P	Lo	Ly	I
	10-8-7	10-8-7	10-8-6	10-8-6
5 UP	E	A	D	G
	D	M	I	Ly
	8-10-11	8-10-12	8-10-12	8-10-12
6 DOWN	G	D	A	E
	M	D	A	P
	14-12-10	13-12-10	13-12-10	13-11-10

A MAJOR A B C# D E F# G#

1 UP	E	A	D	G
	M	I	Ly	Lo
	0-2-4	0-2-4	0-2-4	1-2-4
2 DOWN	G	D	A	E
	I	M	D	A
	6-4-2	6-4-2	5-4-2	5-4-2
3 UP	E	A	D	G
	Lo	P	A	D
	4-5-7	4-5-7	4-6-7	4-6-7
4 DOWN	G	D	A	E
	P	Lo	Ly	I
	9-7-6	9-7-6	9-7-5	9-7-5
5 UP	E	A	D	G
	D	M	I	Ly
	7-9-10	7-9-11	7-9-11	7-9-11
6 DOWN	G	D	A	E
	M	D	A	P
	13-11-9	12-11-9	12-11-9	12-10-9

EB MAJOR Eb F G Ab Bb C D

1 UP E A D G
D M I Ly
1-3-4 1-3-5 1-3-5 1-3-5

2 DOWN G D A E
M D A P
7-5-3 6-5-3 6-5-3 6-4-3

3 UP E A D G
Ly Lo P A
4-6-8 5-6-8 5-6-8 5-7-8

4 DOWN G D A E
Lo Ly I M
10-8-7 10-8-6 10-8-6 10-8-6

5 UP E A D G
A D M I
8-10-11 8-10-11 8-10-12 8-10-12

6 DOWN G D A E
D A P Lo
13-12-10 13-12-10 13-11-10 13-11-10

APPENDIX E — TECHNIQUE REFERENCE

The Beast is unusually demanding on two things that scales normally hide: **muting** and **hand shifting**. Both are invisible in tablature, which is why they are the two things that quietly wreck an otherwise correct practice month.

E.1 Muting

You cross all four strings, repeatedly, in both directions. Every crossing leaves a string behind that must go silent. Three mechanisms, used together:

Plucking-hand floating thumb. The thumb rests on the string below the one you are playing and moves with you. On the E string it rests on the pickup; on the A string it rests on the E; on the D string it rests on the A. This handles everything below you.

Plucking-finger follow-through. When you pluck a string, the finger comes to rest on the next string down. That is not an accident of technique, it is the mute.

Fretting-hand light contact. The fingers not currently fretting lie lightly across the strings above. This handles everything higher than where you are.

The test: play a full ascending sweep, then stop dead on the last note and listen. If you hear anything except that note, your muting is not working and no amount of extra tempo will fix it.

E.2 The WS-WS problem

WS-WS spans five frets. A one-finger-per-fret hand covers four. This is arithmetic, not weakness.

Region	Approach
Frets 1-4	Shift. Finger 1, then finger 2 or 3, then finger 4, hand travelling with the line.
Frets 5-7	Shift or a light reach, depending on your hand. Test with the wrist, not the ego.
Frets 7-12	1-3-4 usually reaches comfortably.
Frets 12+	1-3-4 is easy. Use this region to learn the shape, then take it lower.

The shift is a technique, not a failure. Learn it deliberately: thumb releases, forearm moves, finger 1 lands on the new fret, hand arrives as one unit. Practised properly a shift is inaudible.

E.3 Fretting-hand checklist

Run through this once at the start of every session and once whenever something feels wrong.

- [] Wrist close to neutral, not sharply bent
- [] Thumb behind the neck, roughly opposite fingers 1 and 2, not clamped
- [] Fingers curved, fretting just behind the fret wire
- [] Only enough pressure to make the note speak, released between notes
- [] Shoulder down, elbow hanging, not lifted
- [] Bass positioned so the fretting hand reaches without leaning

E.4 Plucking-hand consistency

Alternate index and middle strictly. The Beast is twelve notes across four strings, which means the “wrong” finger arrives at each crossing on alternate sweeps — that is exactly what you want to train. If you find yourself repeating a finger at crossings, slow down and fix it now, because at speed it becomes a permanent limp.

Raking (letting one finger sweep down across two strings) is efficient and legitimate, but learn strict alternation first. You cannot choose to rake if you cannot not rake.

E.5 When to stop

Stop the session immediately on:

- Sharp pain in hand, wrist or forearm
- Tingling or numbness in any finger
- A wrist that has to bend hard to reach
- Pressure heavy enough to mark the neck

None of these are things to push through. All of them are signals that a shift is being substituted for a stretch, or that the instrument is positioned wrongly. Fix the cause, take a day, come back.

APPENDIX F — PRINTABLE LOGS

Six blank daily logs per page, five pages. Print as many as you need. The log is where the programme becomes self-correcting.

DAY ____ DATE _____ KEY _____ FRET RANGE _____
Top rung reached ____ Clean reps at it ____
Error codes _____
Worst moment _____
Naming score ____/____ Time to first note ____ sec
Hands / wrist _____
Fix tomorrow _____

DAY ____ DATE _____ KEY _____ FRET RANGE _____
Top rung reached ____ Clean reps at it ____
Error codes _____
Worst moment _____
Naming score ____/____ Time to first note ____ sec
Hands / wrist _____
Fix tomorrow _____

DAY ____ DATE _____ KEY _____ FRET RANGE _____
Top rung reached ____ Clean reps at it ____
Error codes _____
Worst moment _____
Naming score ____/____ Time to first note ____ sec
Hands / wrist _____
Fix tomorrow _____

DAY ____ DATE _____ KEY _____ FRET RANGE _____
Top rung reached ____ Clean reps at it ____
Error codes _____
Worst moment _____
Naming score ____/____ Time to first note ____ sec
Hands / wrist _____
Fix tomorrow _____

DAY ____ DATE _____ KEY _____ FRET RANGE _____
Top rung reached ____ Clean reps at it ____
Error codes _____
Worst moment _____
Naming score ____/____ Time to first note ____ sec
Hands / wrist _____
Fix tomorrow _____

DAY ____ DATE _____ KEY _____ FRET RANGE _____
Top rung reached ____ Clean reps at it ____
Error codes _____
Worst moment _____
Naming score ____/____ Time to first note ____ sec
Hands / wrist _____
Fix tomorrow _____

DAY ____ DATE _____ KEY _____ FRET RANGE _____
Top rung reached ____ Clean reps at it ____
Error codes _____
Worst moment _____
Naming score ____/____ Time to first note ____ sec
Hands / wrist _____
Fix tomorrow _____

DAY ____ DATE _____ KEY _____ FRET RANGE _____
Top rung reached ____ Clean reps at it ____
Error codes _____
Worst moment _____
Naming score ____/____ Time to first note ____ sec
Hands / wrist _____
Fix tomorrow _____

DAY ____ DATE _____ KEY _____ FRET RANGE _____
Top rung reached ____ Clean reps at it ____
Error codes _____
Worst moment _____
Naming score ____/____ Time to first note ____ sec
Hands / wrist _____
Fix tomorrow _____

DAY ____ DATE _____ KEY _____ FRET RANGE _____
Top rung reached ____ Clean reps at it ____
Error codes _____
Worst moment _____
Naming score ____/____ Time to first note ____ sec
Hands / wrist _____
Fix tomorrow _____

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Error codes _____
Worst moment _____
Naming score ____/____ Time to first note ____ sec
Hands / wrist _____
Fix tomorrow _____

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Error codes _____
Worst moment _____
Naming score ____/____ Time to first note ____ sec
Hands / wrist _____
Fix tomorrow _____

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Error codes _____
Worst moment _____
Naming score ____/____ Time to first note ____ sec
Hands / wrist _____
Fix tomorrow _____

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Hands / wrist _____
Fix tomorrow _____

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Error codes _____
Worst moment _____
Naming score ____/____ Time to first note ____ sec
Hands / wrist _____
Fix tomorrow _____

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Error codes _____
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Naming score ____/____ Time to first note ____ sec
Hands / wrist _____
Fix tomorrow _____

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Error codes _____
Worst moment _____
Naming score ____/____ Time to first note ____ sec
Hands / wrist _____
Fix tomorrow _____

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Top rung reached ____ Clean reps at it ____
Error codes _____
Worst moment _____
Naming score ____/____ Time to first note ____ sec
Hands / wrist _____
Fix tomorrow _____

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Error codes _____
Worst moment _____
Naming score ____/____ Time to first note ____ sec
Hands / wrist _____
Fix tomorrow _____

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Error codes _____
Worst moment _____
Naming score ____/____ Time to first note ____ sec
Hands / wrist _____
Fix tomorrow _____

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Worst moment _____
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Hands / wrist _____
Fix tomorrow _____

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Error codes _____
Worst moment _____
Naming score ____/____ Time to first note ____ sec
Hands / wrist _____
Fix tomorrow _____

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Fix tomorrow _____

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Error codes _____
Worst moment _____
Naming score ____/____ Time to first note ____ sec
Hands / wrist _____
Fix tomorrow _____

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Naming score ____/____ Time to first note ____ sec
Hands / wrist _____
Fix tomorrow _____

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Top rung reached ____ Clean reps at it ____
Error codes _____
Worst moment _____
Naming score ____/____ Time to first note ____ sec
Hands / wrist _____
Fix tomorrow _____

DAY ____ DATE _____ KEY _____ FRET RANGE _____
Top rung reached ____ Clean reps at it ____
Error codes _____
Worst moment _____
Naming score ____/____ Time to first note ____ sec
Hands / wrist _____
Fix tomorrow _____

Weekly review sheets

WEEK ____ DAYS _____ SESSIONS COMPLETED ____ / 7

Most repeated error code _____
Phase standard passed? _____
Weakest key _____
Slowest string to name _____
Tempo rung: start ____ end ____
Generating or recalling? _____
Adjustment for next week _____

WEEK ____ DAYS ____ SESSIONS COMPLETED ____ / 7

Most repeated error code _____
Phase standard passed? _____
Weakest key _____
Slowest string to name _____
Tempo rung: start ____ end ____
Generating or recalling? _____
Adjustment for next week _____

WEEK ____ DAYS ____ SESSIONS COMPLETED ____ / 7

Most repeated error code _____
Phase standard passed? _____
Weakest key _____
Slowest string to name _____
Tempo rung: start ____ end ____
Generating or recalling? _____
Adjustment for next week _____

WEEK ____ DAYS ____ SESSIONS COMPLETED ____ / 7

Most repeated error code _____
Phase standard passed? _____
Weakest key _____
Slowest string to name _____
Tempo rung: start ____ end ____
Generating or recalling? _____
Adjustment for next week _____

WEEK ____ DAYS ____ SESSIONS COMPLETED ____ / 7

Most repeated error code _____
Phase standard passed? _____
Weakest key _____
Slowest string to name _____
Tempo rung: start ____ end ____
Generating or recalling? _____
Adjustment for next week _____

Final assessment sheet

DAY 30 ASSESSMENT DATE _____

1 Fretboard recall	30 coordinates, 28 correct, under 2 sec	☐ pass	☐ fail	____/30
2 Shape recall	3 shapes from a called fret and string	☐ pass	☐ fail	____/20
3 MILLPAD recall	both directions, any starting label	☐ pass	☐ fail	
4 Row generation	labels, frets and shift, under 10 sec	☐ pass	☐ fail	____ sec
5 Full traversal	C major, open to fret 12 and back, rung 6	☐ pass	☐ fail	rung ____
6 Transposition	any learned key, four sweeps, rung 4	☐ pass	☐ fail	key ____
7 Free start	drawn key and note, within 6 sec, x10	☐ pass	☐ fail	____/10
8 State recovery	halted at random, four variables stated	☐ pass	☐ fail	____/5

TOTAL ____ / 8

Next month's project _____